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Commander
U.S. Army Dugway Proving Ground
ATTN: STEDP-WD-L
Dugway, UT 84022-5000

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Commander
U.S. Army Test and Evaluation Command
ATTN: AMSTE-TM-T
Aberdeen Proving Ground, MD 21005-5055

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13. ABSTRACT *(Maximum 200 words)*

This TOP outlines the procedures, facilities, instrumentations, simulants, and agents of biologic origin (ABO) used at U.S. Army Dugway Proving Ground in testing point biological warfare agent detection systems and components. The TOP applies to field and laboratory testing at all stages of detection equipment development. The TOP addresses quality control and data management issues. It includes recommended formats of data presentation.

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simulant, biosafety, biological warfare agents, aerosol detectors, slit sampler, all-glass impinger

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U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

* Test Operations Procedure (TOP) 8-2-066 15 April 1997
AD No. A328000

BIOLOGICAL DETECTORS, AEROSOL (BDA)

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1. SCOPE.

1.1 Purpose.

This Test Operation Procedure (TOP) provides procedures for developmental and acceptance testing of point biological detectors, aerosol (BDA) for aerosolized agents of biological origin (ABO).

1.2 Application.

a. Procedures will apply to laboratory testing and field testing and to correlation of agent and simulant or tracer testing, as required, with the understanding that field testing will be simulant testing and that correlation will rely on simulant and agent testing comparisons made in a laboratory test chamber. The test director (TD) will need the applicable Test and Evaluation Master Plan (TEMP) or other documents appropriate for the service or country involved.

b. Test procedures will be directed to testing BDA at the subassembly level as a single unit system and as an integrated unit in a larger defined system.

c. The TD will be responsible for all phases of planning, coordinating, and reporting tests and will follow the guidance in the Biological Protection Testing (SEAMLESS) Test Performance Manual (STPM)[1]** . The STPM is not a binding document for testing outside the Department of the Army (DA) but can be used as quality control (QC) guidance by other parties.

1.3 Limitations.

a. Some of the test procedures described will not be applicable to all configurations of BDA. Any test procedures developed must consider the mission and configuration of the pertinent BDA and components.

b. The procedures presented here and the documents referenced may not be applicable as advances in technology facilitate development of new detection capabilities and test evaluation procedures. Frequent revisions to this document will reflect such changes.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

Item	Requirement/Description
a. Field test range.	The primary test ranges for United States Department of Defense BDA testing are Tower Grid and Target S at WDTC. Field tests are appropriate for testing triggering devices and integrated systems. The test ranges were designed to allow test items to operate and perform all mission-essential functions and tasks that are required to accomplish a typical mission profile.

1** Superscript numbers/letters correspond to those in Appendix E, References.

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b. Test facility: Biotron, a safety cabinet for housing and containing ABO aerosol generators, aerosol mixers, primary containment chambers, and referee samplers for testing BDA sensitivity and operation. It is located in Building 2028, Baker Laboratory, WDTC.	The Biotron is a Class III biological safety cabinet that serves as the secondary physical containment barrier for the pathogen testing suite. The Biotron is constructed of stainless steel and equipped with an attached autoclave, glove ports, electrical outlets, lighting, connections for a hydrogen peroxide vapor generator, and a disinfectant-filled dunk tank passthrough. The Biotron is 4 by 4 by 8 ft and has viewing windows for interior observation. The Biotron maintains certification at Class III cabinet specifications (holds 2.0 in of H_2O pressure with no more than 10 percent drop for 30 minutes). All test air leaving the Biotron is double exhaust high-efficiency particulate air (HEPA) filtered and uses four HEPA filters before being introduced into the building air-exhaust system. Test facilities inside the chamber are adaptable to the device tested and required support equipment. The Biotron is equipped with passthroughs for the appropriate air, vacuum, and electrical lines and has the capability of conditioning 500 liters per minute (lpm) of secondary test air with respect to temperature and relative humidity (RH) (4.0 to 40.0°C and 15 to 95 percent RH, respectively). ABO testing will take place under ambient laboratory conditions.
c. Test fixture: Primary containment chamber for testing BDA with ABO.	Test item chambers (TIC) are housed inside the Biotron and constructed of Plexiglas® for direct observation of the test operation. These TICs are designed so that they are approximately three to four times the volume of the BDA test item. A mixing tube receiving HEPA-filtered air is attached to the TIC at the level the test aerosol is expected to be introduced into the test device (the height of the aerosol impactor or analogous air collector). The TIC design includes isokinetic sampling ports for aerosol characterization, viability sampling, and environmental monitoring (temperature and RH). Mixing tubes have connections for aerosol generating devices, secondary test air input, and environmental monitoring. The TIC connects to two high-capacity HEPA filters before being exhausted through the Biotron's air exhaust system.
d. Test facility: aerosol simulant exposure chamber (ASEC), Building 2028.	The ASEC, Room 28, has floor space of 28.5 m^2 and a volume of 78 m^3. The ASEC has ultraviolet (UV) lights to assist in surface decontamination. Both air inlet and exhaust are HEPA filtered with a booster fan on the exhaust system to decrease the time required for appropriate air exchanges. This room is supplied with

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	fans to facilitate the even spread of aerosolized simulant inside the chamber. The inside of the chamber has sufficient air, vacuum, electrical, mechanical, and electronic fittings appropriate for the devices tested and supporting instrumentation. The chamber has windows for observing the interior of the challenge area. Facilities outside the test chamber include a vacuum source sufficient to operate any externally mounted samplers, bench space for desk top computer systems, samplers, Aerodynamic Particle Sizer® (APS®), and other instruments. The room is supplied with a misting RH controller and an indicator and heater with a temperature controller and monitor for operation at ambient levels and warmer.
e. Test facility: Class II biological safety cabinets for containing ABO solutions for testing biological detector or identifier sensitivity and operation, Building 2028, WDTC.	Class II biological safety cabinets serve as the primary physical containment barrier for testing with ABO solutions. The cabinets are constructed of stainless steel and equipped with electrical outlets and lighting. Temperature and RH sensors are mounted as required. These cabinets are certified to Class II cabinet specifications (holds 2.0 in of H_2O pressure with no more than 10 percent drop for 30 minutes). These cabinets are located within equipped BL-3 microbiology support laboratories (Rooms 14, 19, and 76) for assay and preparation of test ABOs. ABO testing takes place under ambient conditions.

2.2 Instrumentation.

Devices for Measuring	Permissible Error of Measurement
Mass	±0.1 mg (for the most sensitive balance used)
Length	±0.1 mm
Time	±1 second
Temperature	±0.5°C
Wind speed	±0.5 m/sec
RH	±5 percent
Wind direction	±2.0 degrees
Aerosol particle number	1 particle per liter (sensitivity)
Aerosol particle size	1 µm (sensitivity)

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2.2.1 Biological Test Instrumentation.

Devices for	Requirement
a. Characterization of challenge aerosols by aerosol particle counting.	An APS® laser particle counter will be used to monitor aerosol particle size and concentration (from 6 x 10 to 1 x 10^6 particles per liter of air). All sample lines will be isokinetic and sample tubing (Tygon®) lengths (not to exceed 2 m) will be shortened to minimize particle loss. For field tests, the APS® provides information on cloud arrival, duration, and departure and will be collocated with the slit and all-glass impinger (AGI-30) samplers.
b. Collecting and characterizing bacterial challenge aerosols by slit sampler.	Slit samplers will be used as the primary aerosol sampler for viable bacteria. The default slit sampler is the New Brunswick Scientific Corporation, (Edison, New Jersey) Model STA-203. This model requires 120 VAC, a vacuum source, and 150-mm diameter bacteriological plates containing an appropriate sterile medium.
c. Collecting and characterizing biological challenge aerosols by AGI-30 (Ace Glass, Inc.).	The AGI-30 will be used as the secondary collector for viable bacteria and the primary collector for toxins and toxin simulants (e.g., ovalbumin) and viruses (e.g., bacteriophage MS2). The nominal flow rate of the AGI-30 is 12 *l*pm. The tip of the capillary stem is 30 mm from the flask bottom. During use, each AGI-30 is attached to a vacuum source that exceeds 15 inches of Hg. The flask will be filled with 20.0 m*l* of appropriate diluent. Sample lines will be placed at the appropriate locations near test equipment (approximately 1 m above floor level in the test chamber through the sampling passthrough to the AGI-30s located outside the chamber). For field trials, AGI-30 will be collocated with slit samplers and sequenced in the same manner.
d. Generating and detecting propylene tracer gas.	Propylene gas cylinders will be collocated with the simulant aerosol generator(s). A real-time propylene sensor array will be used to detect the presence of the aerosol tracer gas in front of and at the sampler line. At the sampler line, propylene sensors will be collocated with the slit and AGI-30 samplers.
e. Disseminating simulant aerosol challenge for a field test of trigger or system sensitivity.	The Micronair Model AU7000 is the default field aerosol generator. This generator can be mounted as a backpack or as a series of individual generators on a mobile platform at the desired height. The size of the generated particles have an mass median aerodynamic diameter (MMAD) of 50 to 60 microns. These particles dry down to an MMAD of 6 to 7 microns 800 meters downwind at the sampling line.

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Devices for	Requirement
	Contract aircraft may be used to disseminate dry powdered simulant through a gravity-fed hopper.
f. Interferant testing.	Interferants will be generated outdoors. Triggers, collectors, and systems will be exposed directly to the interferant. A simulant may be simultaneously released. Outdoor interferant challenges require the same instrumentation as for other field trials. The basic list of battlefield interferants for BDA testing is as follows: fog oil, red smoke from M18 smoke grenade(s), burning gasoline-diesel mixture (1:1), burning polyester and cotton rags, and burning brush (creosote brush and sagebrush). Detector and identifier modules may be tested in the laboratory with liquids from appropriate collectors exposed to field-generated interferants. Known concentrations of simulant or ABO may be added to these fluids.
g. Assay of ABOs and simulants by microscopes, manual and automatic colony counters, balances, chromatography, and spectrophotometers.	Analytical instruments will be used to quantitate the number of colony-forming units (CFU), plaque-forming units (PFU), and/or mass per sample. The coefficient of variance between replicate determinations will not exceed 35 percent.
h. Meteorological measurements.	Meteorological sensors will be located such as to characterize conditions at the sampler line and to facilitate lining up mobile disseminators and detector systems. The locations will vary depending upon test conditions and mobile meteorological stations may be required.

2.2.2 Other Test Instrumentation.

Item	Requirement
a. Still and video camera.	Capable of photographing test items and documenting test events and procedures.
b. Data collection and processing system.	Capable of collecting data on a real-time basis at the frequency and precision necessary for data sufficiency.

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3. REQUIRED TEST CONDITIONS.

3.1 Experimental Design.

The STPM will provide a structure or outline for testing all configurations of the BDA of interest. Using the STPM, the TD will determine the order and sequence for testing to ensure that the testing is performed in a balanced design to support the requirements of this document (e.g., not in a sequential or biased order). A proper balance is essential to achieve adequate data to support the analysis and assessment process.

3.2 Criteria.

Appropriate test criteria shall be identified from the TEMP, Operational Requirements Document (ORD), Independent Assessment Plan (IAP) or other requirements and mission scenarios of the BDA under evaluation.

3.3 Documentation.

The required documentation is described in the STPM.

3.4 Project Log Book, Project File, and Checklist.

The TD shall establish and maintain a project log book, project file, and checklist as described in U.S. Army Test and Evaluation Command (TECOM) Pamphlet 70-3, Project Engineers handbook[2], and in accordance with (IAW) the STPM.

3.5 Development of Abbreviated Test Plan (ATP) or Detailed Test Plan (DTP).

The test planning phase includes identifying potential problem areas by reviewing previous records and the results of similar tests. Relevant Standing Operating Procedures (SOPs) and other procedures to be used shall be reviewed for their applicability, completeness, and adequacy for testing. The development of a test plan requires review of the applicable IAP, TEMP, ORD, STPM and other test guidance literature, familiarization with preceding development and test phases, study of test criteria, and selection of appropriate samples, methods, sequences, facilities, and test equipment. The test plan will include a mission definition as described by a Mission Need Statement (MNS), ORD, or TEMP. The test plan will also include a Data Management Plan (as an Appendix).
The test plan may contain failure definitions. Data from previous similar tests will be considered in order to avoid duplication and reduce the scope of further testing. The ATP/DTP format is described in TECOM Pam 73-1, Technical Test and Assessment Guide[3].

3.6 Safety.

a. Safety Assessment Report (SAR). Before a test is started, the TD will ensure that an adequate SAR has been provided by the developer(s) and that it is understood by the testers. The Installation Safety Officer will review the SAR to ensure that it includes information concerning operational limitations and hazards peculiar to the test item. Testing of an individual component or system will not be initiated until the SAR or equivalent documentation is provided and accepted by the Installation Safety Officer.

b. Procedures. During all test planning, safety concerns will be considered and identified in the SAR review. Safety procedures shall be established by the TD to ensure test operations are conducted so that test participants and nearby persons, property, and equipment are not exposed to safety or health hazards beyond known operational hazards. Applicable safety regulations, safety directives, and SOPs concerning the test item will be followed and all safety requirements will be met. Any individual observing a safety problem shall immediately notify the TD or the

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safety officer and request resolution of the problem. Any individual observing a hazard that places personnel, equipment, or the environment in immediate jeopardy shall take any steps necessary to immediately reduce the hazard and then notify the grid communications network. The Army safety program is described in Army Regulation (AR) 385-10[4].

c. Familiarization. The TD will schedule and ensure that appropriate new equipment training is provided by the developer, as necessary. For combined/joint trials, each equipment sponsor is responsible for ensuring its test personnel meet this requirement. The TD or a designated test officer will provide a safety briefing each day and whenever test participants are changed.

d. Medical Support. U.S. Army Health Clinic Dugway Proving Ground and Ditto Aid Station, WDTC, will provide support to field and laboratory testing. These facilities have SOPs for treating exposure to ABOs. The staff includes emergency medical technicians (EMTs) qualified in advanced life support.

3.7 Environmental Analysis.

In compliance with the National Environmental Policy Act (NEPA) and the 40 Code of Federal Regulations (CFR) 305, Comprehensive Environmental Response, Compensation, and Liability Act[5], the DA requires that a Life Cycle Environmental Assessment be prepared and that potential environmental impacts be assessed at the earliest practicable stage in the planning process of all tests. Proposed testing at TECOM facilities will also be assessed for environmental impact. An appropriate NEPA analysis (e.g., Record of Environmental Consideration (REC)) is required for all tests before testing begins. When a proposed action may significantly affect the quality of the environment, is highly controversial, or is expected to evoke litigation based on environmental issues, an Environmental Assessment (EA) shall be prepared to determine if an Environmental Impact Statement (EIS) is required.

4. TEST PROCEDURES.

4.1 Purpose.

The tests are conducted to obtain data on the functional performance characteristics of a BDA and/or its subassemblies. Appropriate numbers of samples and replications will permit statistical analysis and evaluation of the data.

4.2 Method.

a. The methodology will vary depending on whether a BDA, module, or subassembly is being tested. Three basic types of tests can be performed on a BDA: liquid challenge, aerosol chamber challenge, and field aerosol challenge. The ability of the BDA to return to a baseline state after challenge with a simulant or ABO is to be determined. All aerosol air flows, including test items and samplers used for characterization, will be calibrated before testing.

b. Methodological parameters will also be driven by threat assessments which may include aerosol characteristics including whether the threat is disseminated as powder or liquid. Methodological limitations relative to test requirements must be resolved before testing.

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4.2.1 Liquid Challenge.

a. Some biosensor modules can be tested for sensitivity in the laboratory using liquids prepared in the laboratory or from field aerosol samples. Liquid challenge is the method of choice for detectors and identifiers not being tested as part of an integrated system. These components will be challenged by doses of simulants such as *Bacillus subtilis* var. *niger* (BG) before challenging them with ABOs. This will help establish the relationship between field test results and laboratory ABO results.

b. For screening purposes using materiél developer sensitivity data as a guide, a series of 10-fold dilutions of simulant or ABO that should bracket the device's limit of detection (LOD) will be selected. The highest concentration will be $1X10^7$ CFU/m*l* or 500 ng/m*l* for toxins. This dilution series will be prepared as test samples with one replicate per organism.

c. Examination of the results will permit the TD to select a series of two-fold dilutions. The two-fold dilutions will begin with the lowest of the preceding 10-fold dilutions that elicited a positive response from the detector or identifier. A minimum of eight challenges per simulant or ABO at each of the selected concentrations will be made. In addition, a blank, consisting of diluent or liquid sampler collection fluid, will be tested. These data will be analyzed by weighted regression (probit) to determine the median challenge concentration detected, along with an associated standard deviation.

d. The dose concentration(s) of simulant or ABO will be verified daily by benchmark assays, as appropriate. Labile challenge materials will be used in a timely manner as described in Appendix E.

The operating hours and environmental temperature for each BDA and all maintenance actions will be recorded.

4.2.2 Aerosol Chamber Testing.

a. Chamber testing is appropriate for devices other than triggers that require collection of aerosolized agent. These devices will be tested for relative collection efficiency and safety using simulants. Methodology studies will be performed by the WDTC to identify appropriate baseline referee devices for determination of efficiency.

b. Simulant aerosol challenge will be performed using the large aerosol chamber and instrumentation devices described in Paragraphs 2.1 and 2.2. Aerosols will be generated from slurries or powders. Generated particles will be in the respiratory size range. Aerosol characterization will include MMAD and numeric mean aerodynamic diameter (NMAD). The specific atmospheric environmental conditions and aerosol concentration ranges to be used will be described in the test plan. Aerosol collection efficiency data will be used in deriving dilution factors for the aerosol challenges. Dose concentration range will be described in the requirement documents.

c. ABO testing requires the Class III biosafety cabinet, test fixture chamber, and associated instrumentation described in Paragraphs 2.1 and 2.2. No testing with aerosolized live/active ABOs is conducted without proper EMT support present at the Ditto Aid Station, WDTC. Some ABOs require the presence of two test personnel when any procedure with potential aerosolization is performed (Appendix E). Decontamination procedures appropriate for the BDA and test fixtures will be followed. Appropriate procedures are those for which approved SOPs exist.

d. AGI-30 and/or slit sampler samples will be taken from 1-minute before to 1-minute after each BDA challenge to measure change in the challenge concentration during the trial. The AGI sample will be analyzed in triplicate using methodology appropriate for the BDA being tested.

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4.2.3 Aerosol Field Testing.

a. Field aerosol testing may be conducted on either BDA triggers or integrated systems. While field testing is more likely to produce aerosol challenges representative of threat conditions, such aerosols are inherently difficult to replicate. In addition, challenge aerosol dose delivered on site is difficult to control because of micrometeorological variation.

b. Slit samplers will provide the best estimate of challenge dose. AGI-30 data may be used in a similar manner, but generally will lack the time resolution necessary. Assay of the samples will be similar to that described for liquid and aerosol chamber testing. CFU/liter, PFU/liter, or mass/liter of air will be determined for each simulant challenge. Agent-containing particles per liter of air (ACPLA) may only be calculated directly from slit sampler plates.

c. The minimum meteorological support for a field test of a biological agent detector should consist of the following observations: wind speed and direction measured at a height of 3 m; cloud cover and cloud; present weather; visibility; RH; ground condition; air temperature; solar radiance; change in temperature over time; and lapse conditions during testing. These standard meteorological observations are needed to document the conditions under which the detector is tested. The wind direction and wind speed should be sampled at a rate of 1 Hz to provide at least limited turbulence information for use in any post-trial dispersion model analyses. The cloud cover information is needed to establish the Pasquill stability category, which is a measure of the dispersive state of the atmosphere.

d. The following meteorological constraints will be followed for testing of BDA: no precipitation; a 10 min mean wind speed of 2 to 6 meters per second measured at a height of 3 m; wind direction limitations consistent with the constant concentration widths described above and the ability to move the disseminators; and Pasquill stability category of C, D, E, or F.

e. The BDA collection inlet will be located 3 m above the ground and a minimum of 800 m downwind from line source generators. The generators will release the aerosol at a minimum height of 2 m. The APS® and propylene detectors will be used to determine arrival of the cloud at the sampling site. Slit samplers and AGI-30s will be used to quantify the challenge.

4.2.4 Interferant Studies.

a. Interferant studies are conducted to test for false positive or false negative reactions. The following interferants will be used: fog oil, red smoke from M18 smoke grenade(s), burning gasoline-diesel mixture (1:1), burning polyester and cotton rags, and burning brush (creosote brush and sagebrush). Additional interferants may be used when they are available. Only one collection of each material will be made. Interferants will be generated outdoors. When applicable, interferants will be collected by swab (i.e., from, for example, a just-fired artillery tube or spent munition) and resuspended in the appropriate collection fluid. Triggers, collectors, and systems will be exposed directly to the interferant. A simulant may be simultaneously released or spiked into a collection fluid. Outdoor interferant challenges require the same instrumentation as for other field trials.

b. Detector and identifier modules may be tested in the laboratory with liquids from appropriate collectors exposed to field-generated interferants. Known concentrations of simulant or ABO may be added to these fluids.

4.2.5 Retrograde Operations.

a. Decontamination. BDA will be decontaminated before removal from containment. Where possible, fluidic systems will be flushed with liquid decontaminants appropriate for the ABOs used. Hydrogen peroxide vapor may

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be used where surfaces and components may be sensitive to liquid disinfectants.

b. Disposition of Equipment. The test sponsor will provide information for inclusion in the test plan relative to equipment disposition following test.

5. DATA REQUIRED.

The data required will vary with device but should, as a minimum, include:

a. Characterization of the challenge including concentration, duration, and aerosol parameters with a resolution of approximately 2 seconds for slit samplers and no greater than 1 minute for AGIs;

b. Unit and subassembly identification numbers;

c. Complete documentation of procedures followed, sequences of events, record of observations, deviations, anomalies, and difficulties encountered;

d. Photographic documentation of representative test methods used, equipment and instrumentation systems, irregularities, component failure, or other anomalies, as required, on the device being tested;

e. Environmental conditions (temperature, RH, and, for field tests, wind vectors), at critical test sites throughout the test;

f. Set-up and operation time;

g. Time required to make necessary adjustments and repairs;

h. Time required for unscheduled troubleshooting and maintenance;

i. Continuous recordings of critical system performance parameters;

j. Observations that are relevant to future training requirements, operator requirements, deployment, and logistics support;

k. Instrument calibration records;

l. Screened APS® data files;

m. Certification that all test and referee devices containing clocks are synchronized.

6. PRESENTATION OF DATA.

a. Data presentations address the agreed-upon objectives and supporting criteria of the test and are in terms and units agreed upon through the SEAMLESS test process as referenced in the data management section (Paragraph 7).

b. Data presentations are designed in a format that rapidly convey whether criteria have been met, partially met, or not met. These are appropriate for the main body of reports and briefings. Engineering-oriented graphs are best placed in appendices.

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c. Titles and legends of tables and figures will provide sufficient information for interpretation of the item without reference to the main body of text to include abbreviations and units of measure.

d. It is strongly recommended that all data, graphs, and figures and the formal written test report be published in an interactive compact disk - read only memory (CD-ROM) format. This format may contain raw and reduced data in spreadsheet and database files, as appropriate, to the test. Unless the test sponsor directs otherwise, both referee and test item results will be included.

e. Data acquisition and storage should be accomplished in real-time mode whenever possible. This will minimize transcription errors and data loss and facilitate data reduction and presentation. Also, automatic data reduction "shells" may be built into the system which could provide rapid evaluation of test progress.

f. Evaluation of BDA trigger challenge data should include the following data: propylene cloud profiles, APS®, background slit samplers, sequential slit samplers, AGI, and meteorology. Organize these data onto Gantt charts (Figure 1) and other appropriate graphic displays.

7. DATA MANAGEMENT.

A single individual will be given overall responsibility for data management. A data management plan will be an appendix to the test plan. The TD will coordinate the data management plan with test center statisticians, test team members, the customer, and TECOM, if applicable. The plan should include the following concepts:

7.1 Flow Chart.

The TD will prepare a flow chart for data generated by referee and test subject BDA devices. This chart will include chain-of-custody information, QC and quality assurance (QA) steps, archiving, and data reduction.

7.2 Referee Data.

Aside from verifying proper mechanical function (i.e., flow rates) of a sampler and its provision with any required media, referee devices described below have no calibration or other standards with which they may be challenged. Critical orifices built at DPG are calibrated by the U.S. Army Test Measurement and Diagnostic Equipment (TMDE) support group at DPG. Otherwise, these referee are the methodological standards.

a. Meteorological data will be presented in a format to facilitate a rapid determination that the test occurred within the specified meteorological envelope. The format used on the CD- ROM for Joint Field Trial II was acceptable. Meteorological data will be electronically transmitted from the grid.

b. APS® data are displayed using the appropriate version of Dycor Corporation, Edmonton, Alberta, software. From these displayed data, times may be derived for cloud arrival and departure from the grid. The data displays provide information about approximate background particle concentrations before and after aerosol cloud passage. These data are not a primary source for determining cloud concentration because there is no information on particle composition, but they do include particle size, distribution, and concentration data over time. Means and geometric standard deviations may be calculated from these data. APS® data will be electronically transmitted from the grid.

c. Propylene data, because of the high sensitivity of the detectors (less than 1 ppm), are used primarily to verify the aerosol cloud arrival time, departure time, and distribution across the grid. The concentration data are useful only in a qualitative sense. These data are not limited by distance from the point of generation, but by concentration above background. Propylene detectors generate real-time output used to provide a preparative cue for initiating referee sampling. Propylene data will be electronically transmitted from the grid.

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d. Slit sampler data are the primary data set for quantifying bacterial simulant challenges. Counts from sequential plates for a single position are appended. Each sector count is paired with the time stamp corresponding to the end time of that sector. Sector counts are per volume of air impacted during the sector sweep time. The trial challenge concentration for a single tower or position is the maximum moving average value in ACPLAs during the period the aerosol cloud was moving across that position. These reduced data (in ACPLAs) will be stratified into high (150 to 250 ACPLAs), medium (25 to 149 ACPLAs), and low (5 to 24 ACPLAs) data strata. Very high or very low data strata may also be used if data fall outside the 5 to 250 ACPLA range. In addition, slit sampler data are useful for determining the levels of reaerosolized simulant and aerosolized native bacteria. Slit sampler plates are electronically counted. The sections (Paragraphs 7.3.a and b) suggest specific ways in which slit sampler data should be used relative to those device types. These data must be used consistently for a given device type.

e. AGI data are primary data for determining aerosol concentrations of toxin simulants and virus simulants. These data provide secondary concentration data for bacterial simulants. AGI data do not have the resolution of slit sampler data because the sampling time is usually at least 1-minute. For technical reasons, no method has been proposed for converting AGI units into ACPLAs. When using 1-minute samples, the sampler with the highest value (expressed in mass or counting units per liter of air) is the challenge concentration. AGI data are always expressed as units per liter of air. AGI plates are electronically counted. Definitions of low, medium, and high AGI values have not been defined. AGI data will be present in a similar format to slit sampler data.

f. As they become available, sensitive technologies that provide a complement to viability and activity assays will be used to describe simulant and ABO challenge doses. Such technologies may include, but are not limited to, antibody-based assays, nucleic acid detection and amplification, chromatography and spectroscopy, and variations of light, electron, or atomic force microscopy.

7.3 Biological Detector, Aerosol (BDA) Component Data.

The test customer and the TD must agree before completion of the test plan, whether point estimates or probabilities with a specified level of confidence are required. The type of summary statistics required greatly affect the test format, work load, and analysis of data and would be appropriately developed as part of the SEAMLESS task for developing the test outline. When completed, a more timely and accurate cost estimate is possible. Data requirements for a specific test program may be more or less inclusive than described (Paragraphs 7.3.a, b, c, and d). A test to screen a number of devices for further testing may have different data requirements than a test for product improvement for which many parameters already are known.

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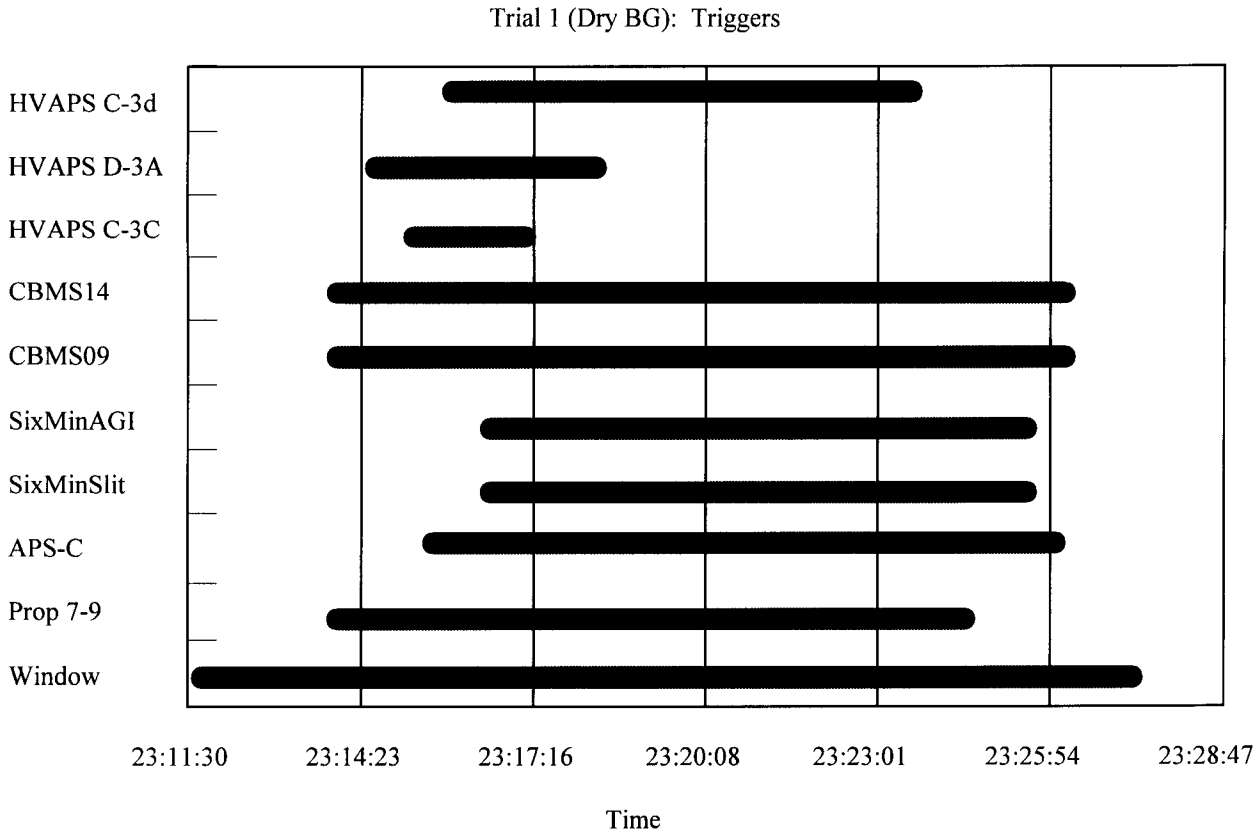


Figure 1. Presentation of Trigger Data in a Gantt Chart.

Note: All labels on the chart will have to be self explanatory to its intended audience.

a. Triggers. Triggers are devices which alert a system to the possible presence of particles of biological origin. Triggers will be evaluated on the field aerosol concentration at which they alerted or the maximum concentration to which they were exposed (as described in Paragraph 7.2.d). The concentration at which a trigger alerts will be determined by subtracting the sample processing time from the alert time and integrating the sector counts (in ACPLAs) across that time period. False alerts can only be evaluated if referee data are available. Triggers may also be evaluated on response time to a concentration stratum. Trigger data are best displayed in a Gantt chart format which allows the trigger alert, or lack thereof, to be compared visually with the pertinent referee data (Figure 1). Figure 1 plots the triggers as a duration of alert time, but a customer may desire to plot only the time of alert. Several variations are possible, but any graphic presentation should be of obvious. Histograms (Figure 2) should also be prepared to summarize probability of detection by concentration strata.

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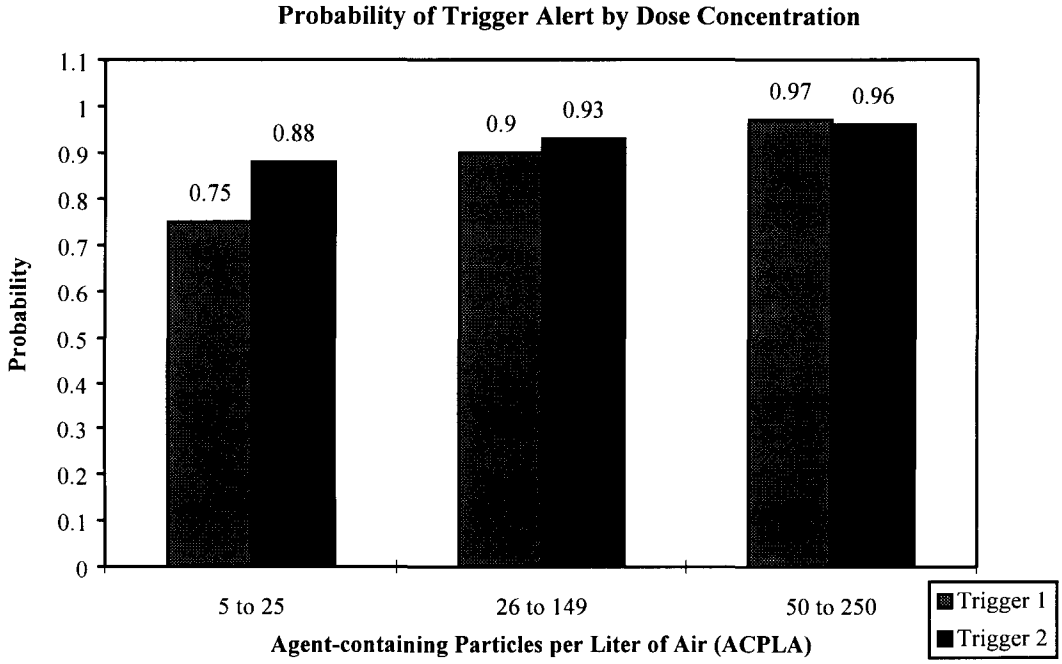


Figure 2. Example of Data Presentation for Two Triggers.

b. Collectors. Collectors are devices that capture and concentrate aerosol particles for analysis by detectors and identifiers. For discussion purposes, this excludes devices whose primary function is to provide particles to a triggering module. Collectors are best tested in the ASEC where they may be challenged with a more-or-less stabile concentration aerosol. Collectors may be evaluated with respect to collection efficiency for particles of different composition and size. A concentration factor may be calculated. An AGI-30 will run concurrently with the collector. The units used (e.g., mass, antibody-related, nucleic acid-related, direct count, plaque or colony) will be appropriate to the agent or simulant. A normalized concentration factor is calculated using the following formula:

Concentration Factor = units/mL $_{\text{collection device fluid}}$ / units/liter $_{\text{air collected by an AGI-30 from aerosol cloud}}$

Collectors evaluated in field tests as integrated detection system components may be evaluated for concentration efficiency using slit sampler or AGI-30 data, as appropriate, for the time period during which collection was accomplished. Figure 3 is an example of a simple histogram prepared from such data. Discussions of collectors should include aerosol cloud characterization data (NMAD, distribution profile, geometric standard deviation, etc.), collector air sampling rate data, sample volume, and data showing whether it was collected continuously, intermittently, or segmentally.

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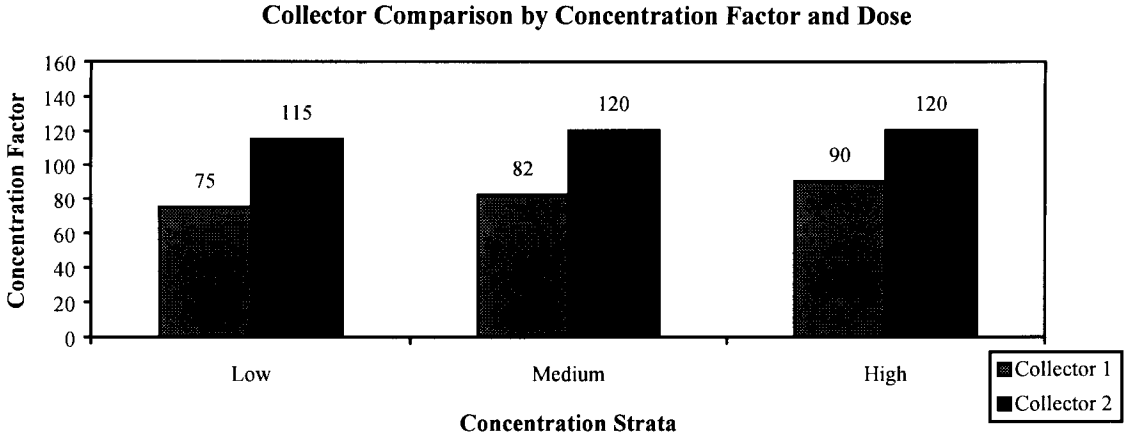


Figure 3. Sample Histogram of Collector Concentration Factors.

c. Detectors. Detectors are devices which provide presumptive evidence of biological material in an aerosol sample. Depending on the device, this evidence may vary from general to specific. Their function may be combined with that of triggers and collectors. Data from detector challenges will be used to calculate probabilities of detection by concentration and by simulant or ABO. Blanks, battlefield interferants, and environmental samples provide the basis for determining false positive rates. Response time will be measured. Unless a test directive specifies otherwise, the following additional data will be collected: time to set up, time to calibrate, and observations on the reliability and maintainability of the devices.

d. Identifiers. Identifiers are devices that function to confirm a presumptive result. Their function may rely on specific antibodies or nonantibody-based technologies. Identifier sensitivity is based in the smallest number of aerosol particles or mass it can identify. Its specificity is based on the number of particles or mass it correctly identifies. Sensitivity and specificity or probabilities of correct identification may be determined for specified concentration strata. Blanks, battlefield interferants, and environmental samples provide the basis for determining false positive rates. Response time will be a measurable variable only if the identifier device has a processing time which is concentration dependent. Some devices may have parallel processing which makes multiple simultaneous determinations possible. Unless a test directive specifies otherwise, the following additional data will be collected: time to set up, time to calibrate, and observations on the reliability and maintainability of the devices. Graphical presentations will be used.

7.4 Definitions.

a. Percent sensitivity = true positives/(true positives + false negatives) x 100

b. Percent specificity = true negatives/(true negatives + false positives) x 100

7.5 Quality Control.

a. Before data are collected, test planning must be subjected to a data quality objective process.

b. Data collection sheets will be presented or designed so that no interpretation is required to transcribe data from an instrument display.

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c. Data terminal input formats, if appropriate, should be designed so that no interpretations or selections are required by the data input technician when transcribing from data collection forms.

d. Data collectors will be used who have been given familiarization training on the device from which they are collecting data.

e. Pretest operational readiness inspections (ORIs) will include data transmittal and reduction. Time will be scheduled to resolve any data flow issues prior to commencement of testing.

f. The test plan will detail sample labeling and chain-of-custody procedures.

g. Data will be reviewed by the TD and data manager for consistency and internal coherence to determine if the data streams make sense with respect to each other or determine if there are data outliers that need to be resolved?

h. Retain under appropriate storage conditions, aliquots of challenge materials and referee samples (i.e., AGI-30 fluids) until all data have been reviewed and further specimen assay is not required. Destruction of the aliquots should be coordinated with the customer.

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APPENDIX A. Instrumentation

Aerodynamic Particle Sizer® (APS®)

The Aerodynamic Particle Sizer® (APS®), Models 3300 and 3310 (Figure A-1), are used to determine aerodynamic diameter (μm), number concentration (particles per cc), mass concentration (mg per m^3), and surface concentration

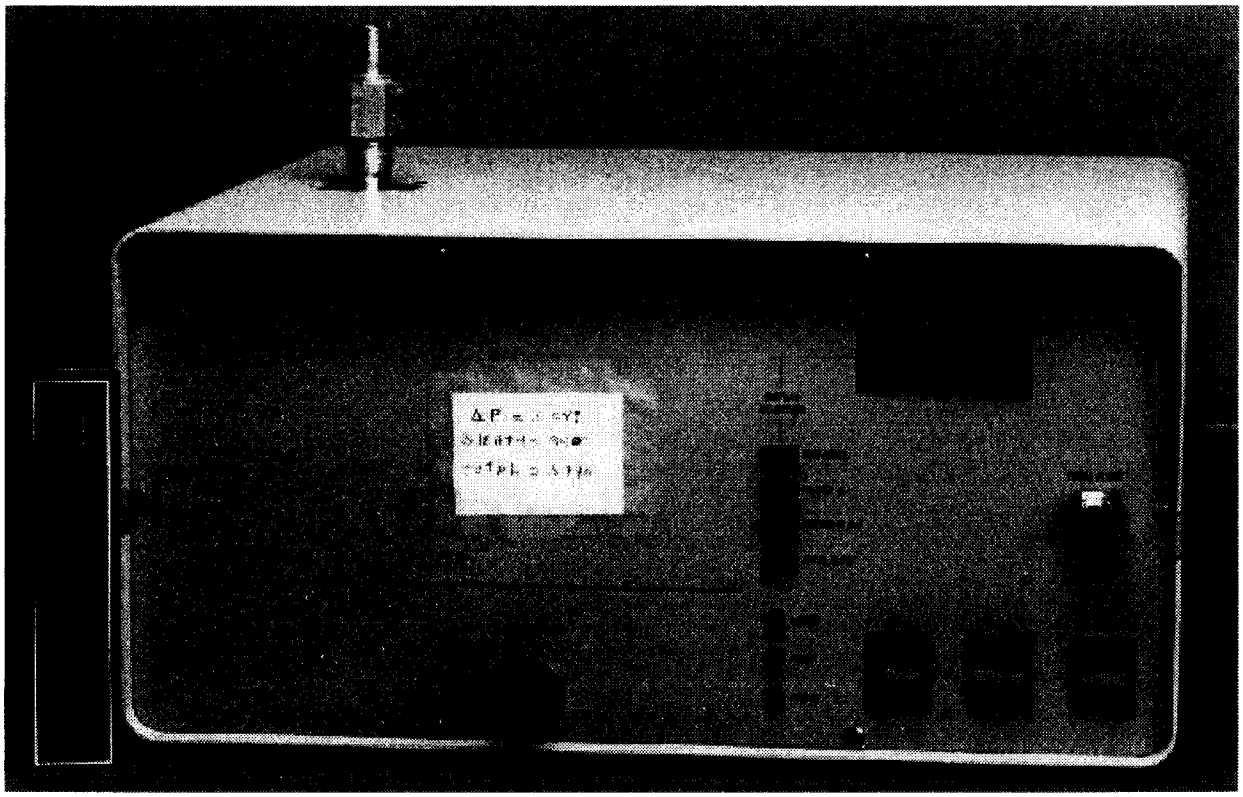


Figure A-1. Aerodynamic Particle Sizer® (APS®)

(μm^2 per cm^3). The APS® samples 5 *l*pm, of which 4 *l* are used to provide clean sheath air and 1 *l* is used for the sample. Procedures for the normal operation and calibration of the APS® are outlined in SOP No. (Pending). Under normal operation, the APS® is calibrated every six months. DPG has observed that calibration is very stable and requires minimal adjustment.

The APS® is used to characterize aerosols outdoors and in chambers. For outdoor field testing, the APS® is used to determine aerodynamic diameter, the arrival of simulant aerosol challenges, and number concentration. The APS® will be operated for the entire test period outdoors. An APS® is used at each referee station during outdoor trials. Data is collected using specially designed Threat Detecting Software (Dycor Corporation, Edmonton, Alberta). This software allows for simultaneous collection of data for up to nine APS® systems. This data is normally transmitted to a computer via freewave radio modems. APS®s are used to characterize both simulant and ABO aerosols in the laboratory. When simulant aerosols are generated, the APS® is used to determine aerodynamic particle size and to follow trends in particle concentration. When the APS® is used in BL-3 containment, it provides aerodynamic particle size and physical particle counts. An APS® has been dedicated to the BL-3 chamber and is routinely decontaminated with vapor hydrogen peroxide. Under normal operating conditions in the chamber and after

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appropriate warm up time, APS® data will be saved to disk 5-minutes before aerosol generation to determine background particle levels. The APS® will operate continuously so that cloud arrival, cloud build up, and maximum concentration values may be recorded. When the test is completed, secondary air is introduced to purge the chamber, and the decrease in particle concentration to background level may be observed.

Biological aerosols are collected through Tygon® tubing (minimum 0.5-in. internal diameter) with very little particle loss. The length of Tygon® tubing should be minimized. It has been demonstrated (methodology study) that BG can be sampled through 40 feet of Tygon® tubing with approximately 5 percent particle loss. Other types of tubing should not be used to sample biological aerosols.

Problems with particle coincidence are significant (10 percent error) at concentrations above 4000 particles per square centimeter. Under normal test conditions these concentrations are rarely generated.

Slit Sampler

the S-T-A Biological Air Sampler Model STA-203 slit samplers (Figure A-2) is used to determine aerosol cloud concentration in ACPLA. Several different clock motors have been installed in the slit samplers for different applications. A series of three 24-minute slit samplers are used to characterize biological background. A series

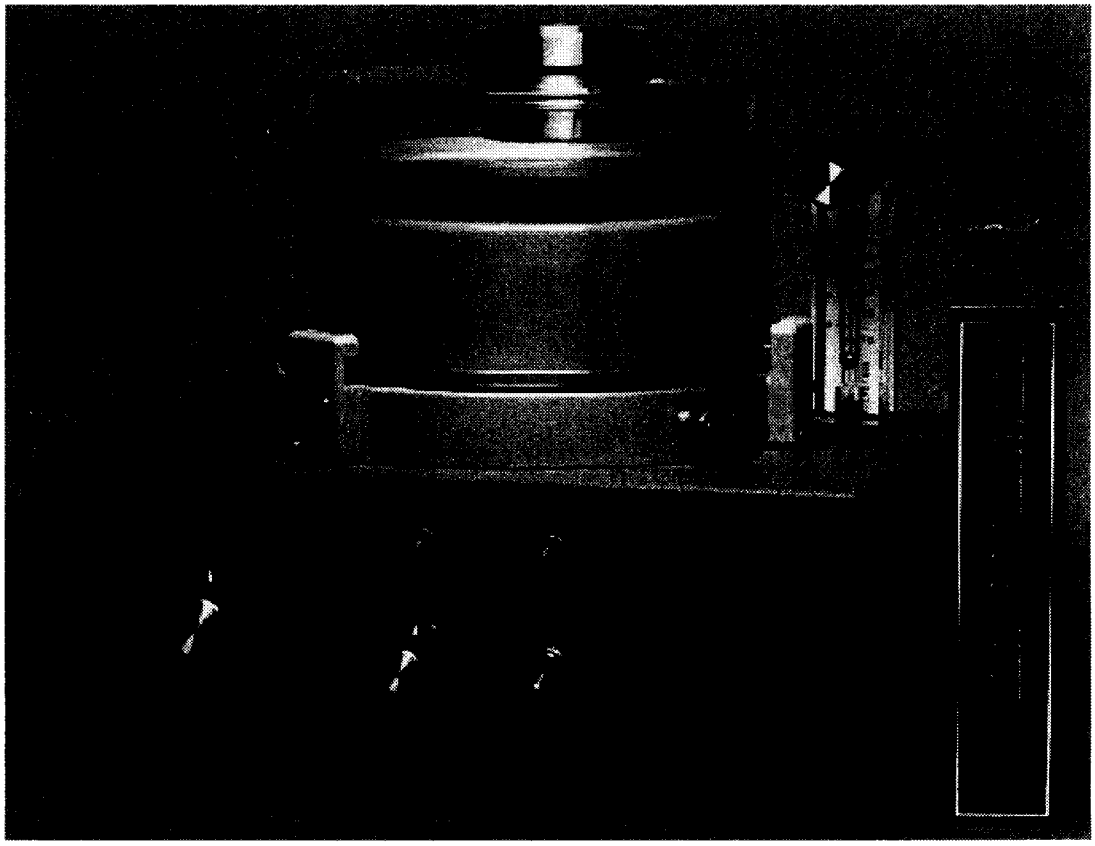


Figure A-2. The New Brunswick Model STA-203 Slit Sampler.

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of six to nine 1-minute slit samplers (sequenced using M37 automatic sequencer) is used to qualitatively characterize challenge aerosols. Additionally, one 10-minute slit sampler is used to qualitatively evaluate the total dose of the challenge aerosol. Background slit samplers operate continuously throughout the test period. The 1-minute and 10-minute slit samplers are turned on at least 1 minute before cloud arrival at the test area (this is determined using propylene sensors located 100 m upwind from referee stations) until respective sampling sequences are completed. The trailing edge of the challenge aerosol may not be fully characterized if the generation period is longer than 10 minutes. All slit samplers will be collocated on their respective referee towers. The 1-minute slit sampler data is broken down into 30 sectors (or a 2-second sample) per 150 mm plate (Figure A-3).

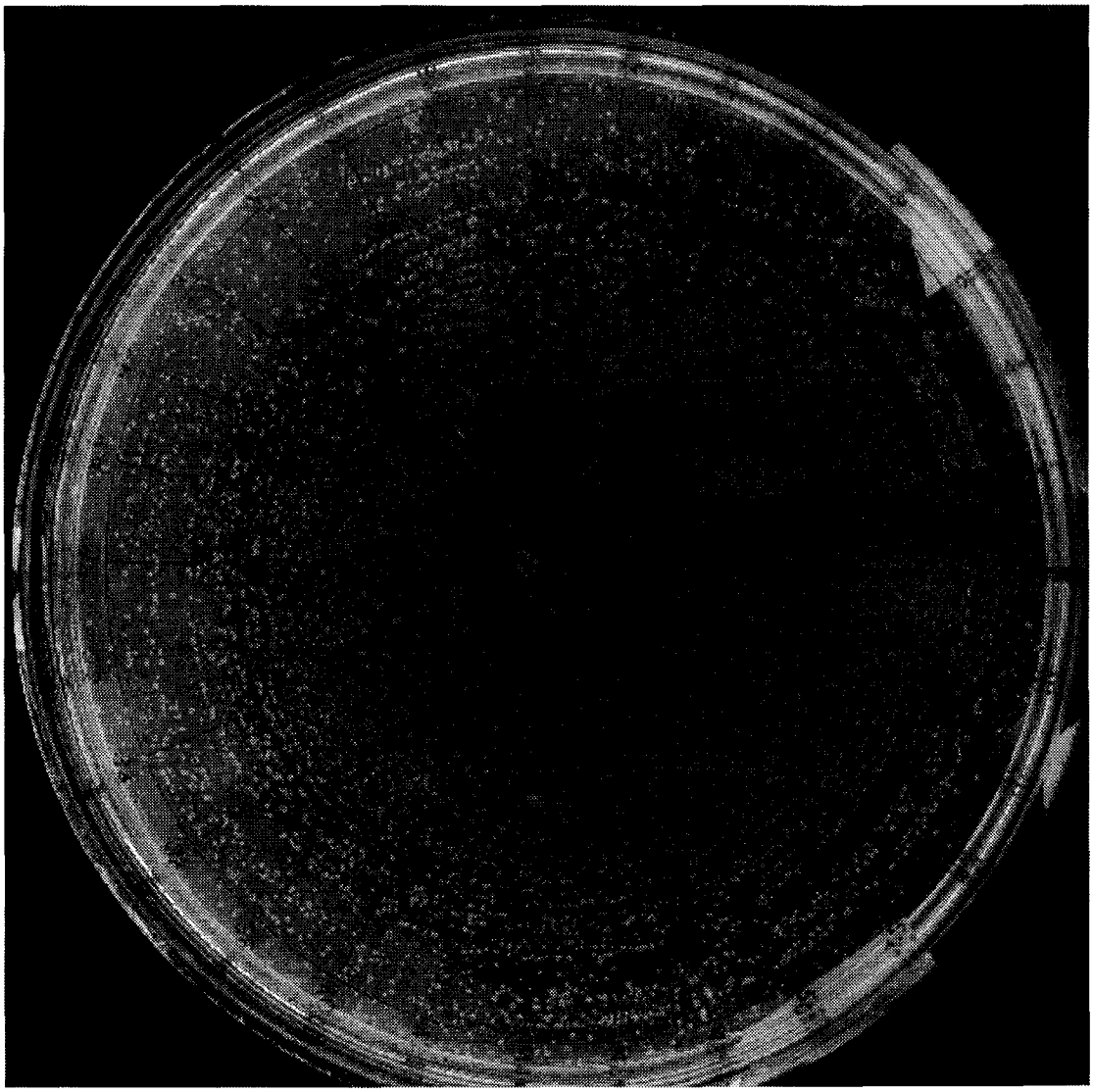


Figure A-3. Slit Sampler Plate, 15 cm, on a Thirty Sector Enumeration Template.

Note: the bacteria are BG collected on trypticase soy agar containing tetrazolium dye.

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Because slit samplers operate at 1 cfm (28.3*l*pm), each sector represents 0.94 *l* of sampled air. Data will be presented sector by sector to provide a single series of data. This technique allows for accurate alignment of ACPLA values with test times.

Slit samplers (1-minute or 10-minute) are used in chamber studies to evaluate ACPLAs. Depending on the length of the trial, slit samplers may be sequenced to cover the test period. When slit samplers are used with ABOs, they should be placed inside a HEPA-filtered Plexiglas® box to contain the pathogen aerosol. Vacuum lines connected to the slit sampler must also be HEPA-filtered. Slit samplers are not normally sequenced during ABO testing but are used to collect a "snapshot" of the challenge aerosol.

Calibration of slit samplers includes timing, the rates of rotation of the sample plate, and a quantitative check of the airflow through the sample line. Airflow calibration can be accomplished using 1 cfm critical orifices. Consumables used for slit samplers include 150 mm plastic plates which contain 65 m*l* of growth media, which is dependent upon the biological simulant being sampled.

Limitations associated with using slit samplers include the following: where challenge concentrations are such that overlap of particles on the agar surface are difficult to quantify, slit samplers are of limited value; and slit samplers collect intact airborne particles so that individual CFUs cannot be enumerated. Additionally, our experience has shown that New Brunswick slit samplers are designed for laboratory use, and care must be taken when they are used outdoors. When temperatures are around 45° F or colder, the agar surface around the aerosol inlet may freeze.

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All-Glass Impinger (AGI-30) (Chemical Corps Type)

General Description: This high velocity impinger (Figure A-4) is a modification of the Greenburg-Smith impinger that was primarily used in air pollution studies. The upper end of the inlet tube is bent into a 90-degree curve of large radius, and the stem and flask are fitted together through a ground glass connection. A short length of capillary tubing is fused to the lower end of the inlet tube so that the tip is 30 mm from the flask bottom. The 30 mm raised stem provides an increase in the collection of viable vegetative cells. The vacuum outlet arrangement is similar to that of the Greenburg-Smith sampler.

Catalog Description, Ace Glass, Inc., Vineland, NJ 08360:

7540 Impinger, Air Sampling

A high velocity impinger (Ref. AGI-30) which passes 12.3 to 12.6 *l*pm (corrected to engineering standard gas conditions) when the pressure drop across the orifice is 41 cm Hg or greater. Tip of capillary stem is 30 mm from the flask bottom which provides an increase in the collection of viable vegetative cells. Joint is TS 24/25. Capacity approximately 125 m*l*.

Optimum Flow Rate: 12.5 *l*pm.

Collection Surface: Liquid (20 m*l*).

Operation. The AGI-30 is subject to freezing under some combinations of sampling time and ambient temperature. Under such conditions, the TD may elect to run the AGI-30s in a heated rack (Figure A-5). Agi-30s may be transported to the test site in specially designed carrying racks (Figure A-6). An M37 Vacuum Sequencer (Figure A-10) may be used when sequential samples are required.

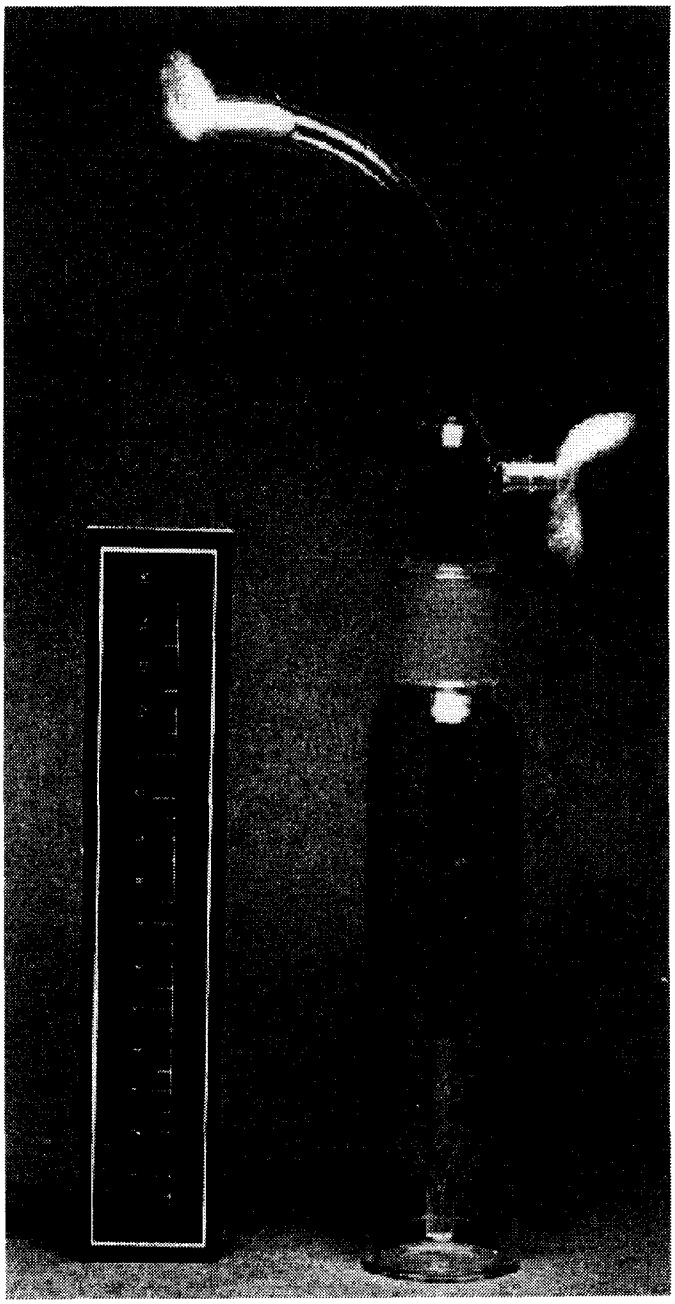


Figure A-4. All-Glass Impinger (AGI-30).

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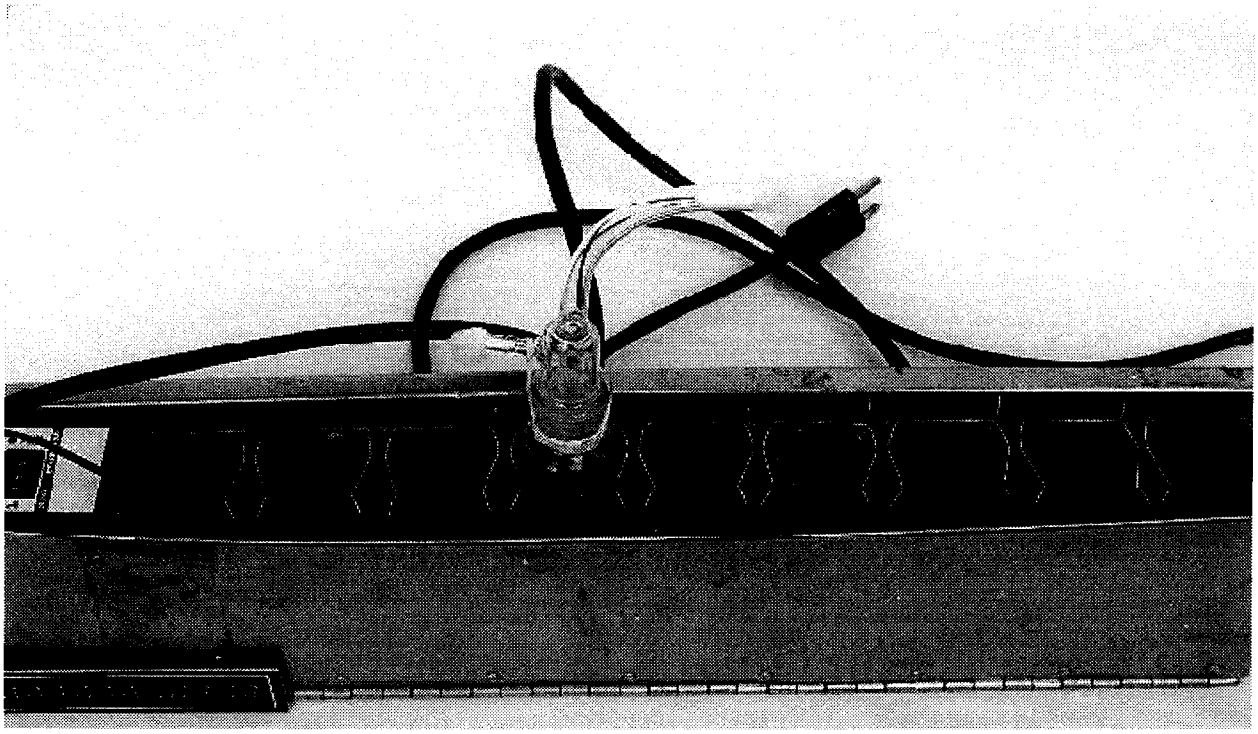


Figure A-5. Heated Sampling Rack for AGI-30s.

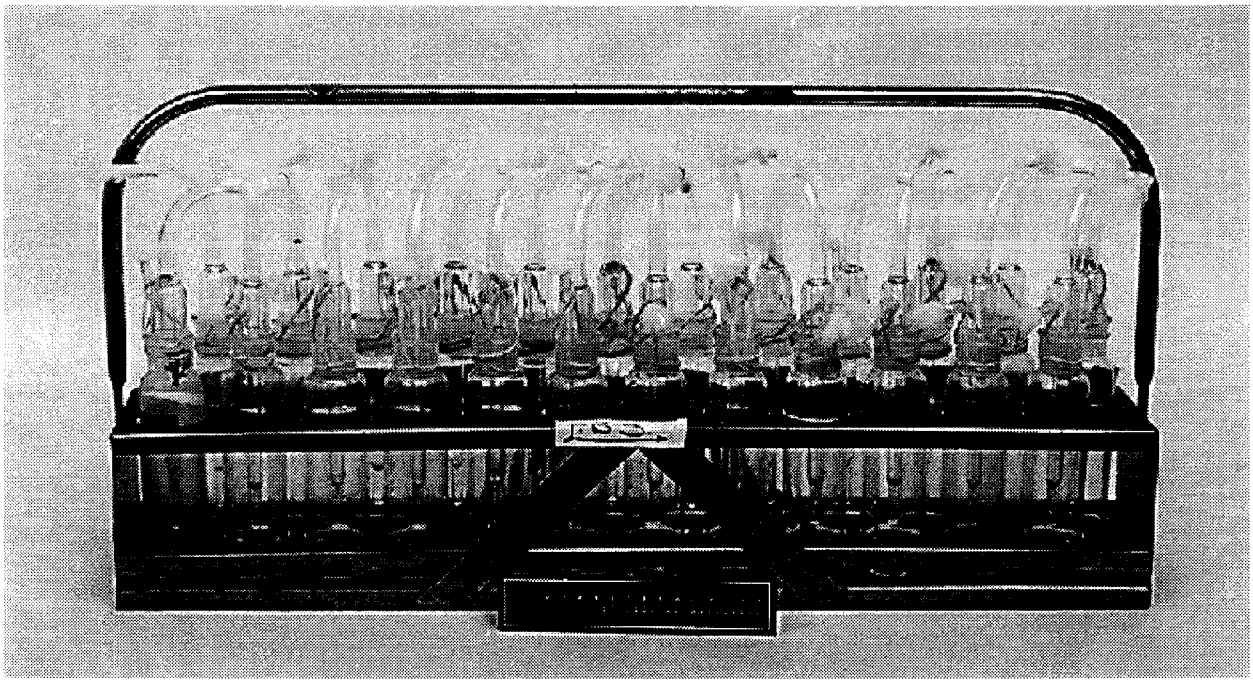


Figure A-6. Field Transport Rack for AGI-30 Samplers.

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Micronair Aerosol Generator

The AU8000 Micronair Agricultural Generator spray head is used to disseminate both wet and dry biological simulants (Figures A-7 and A-8). Simulants that have been aerosolized with the Micronair include both wet and dry BG, wet *Erwina herbicola* (EH), wet bacteriophage MS2, and dry ovalbumin. The Micronair is mounted on a

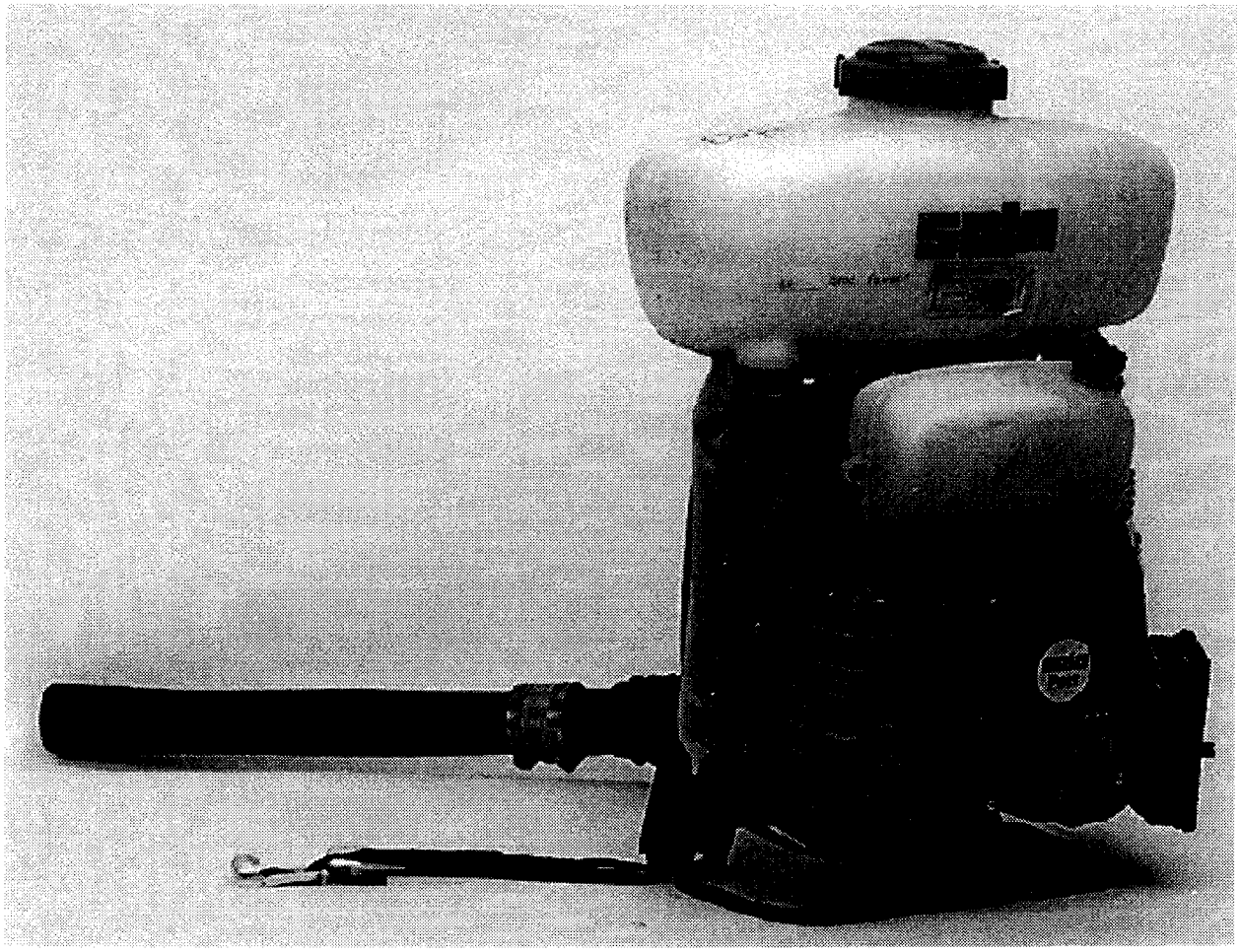


Figure A-7. Micronair Generator with Tube Attached for Dry Simulant Dissemination.

backpack attachment, the length of the spray head is 30 cm, the diameter is 15 cm, the weight is 12.2 Kg, the air inlet tube diameter is 65 mm, and the liquid flow rate is variable from 0.02 to 1.2 *l*pm. The capacity of the slurry tank is 14 *l*. The output of the air blower is 20 cubic m per minute with a velocity of about 125 m per second at the outlet. The spray head rotation is powered by a two-cycle gasoline engine. The size of the droplets produced depends on the rotational speed of the atomizer, the liquid flow rate, and the properties of the slurry. The Micronair Agricultural Generator is designed to produce large droplets, between 50 μm and 400 μm, that settle quickly. To produce the smallest possible droplets (50 μm), The unit is operated at near full throttle with a liquid feed rate of 1 *l*pm. A typical slurry concentration for BG is around $1X10^{10}$ CFU/m*l*. When this slurry is atomized under the above conditions a 50 μm droplet is produced. When the water associated with the droplet totally evaporates the resultant particle is about 5 to 7 μm in diameter. At DPG (because of the low humidities) these droplets dry to equilibrium within a few seconds after generation.

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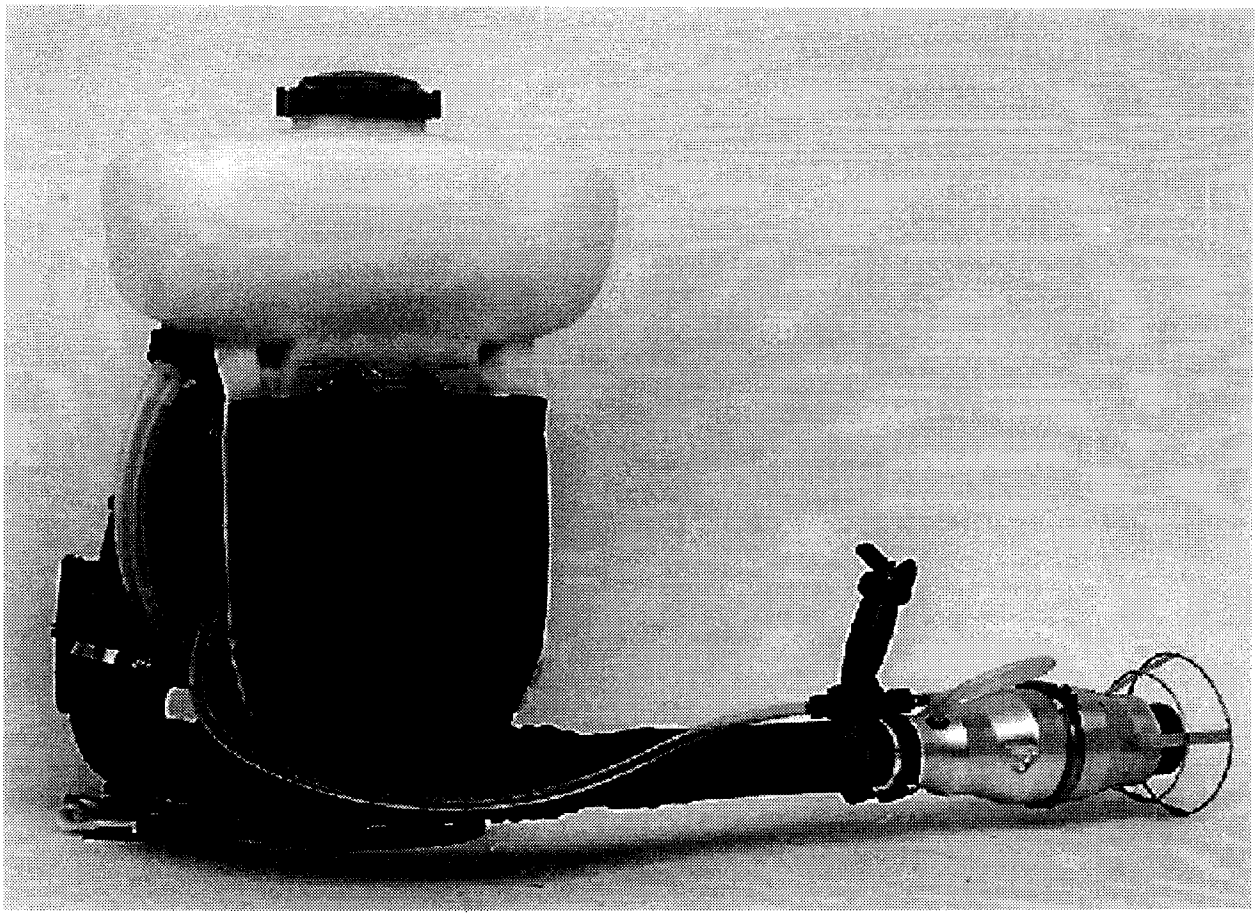


Figure A-8. Micronair Generator Rigged for Disseminating Liquid Simulant Slurry

Calibration of the Micronair Aerosol Generator is restricted to measuring the slurry flow rate. Under normal operation, one to three generators are operated during a given test. When one generator produces particles (BG), that are characterized 800 m away, concentrations measured are 5 to 15 ACPLA. Dry simulants (BG and ovalbumin) must be manually fed into the spray head. Control of the output is difficult under these conditions. During large-scale field tests only one simulant is ever used for a given Micronair. This practice should alleviate problems connected with cross-contamination of simulants. Micronair generators may be mounted as a rank on a mobile platform (Figure A-9).

Limitations associated with use of the Micronair to disseminate biological simulants include lack of control for feeding dry simulants and problems associated with the formation of ice crystals at temperatures at or below 45° F. Overall, the Micronairs have been very dependable during outdoor field trials.

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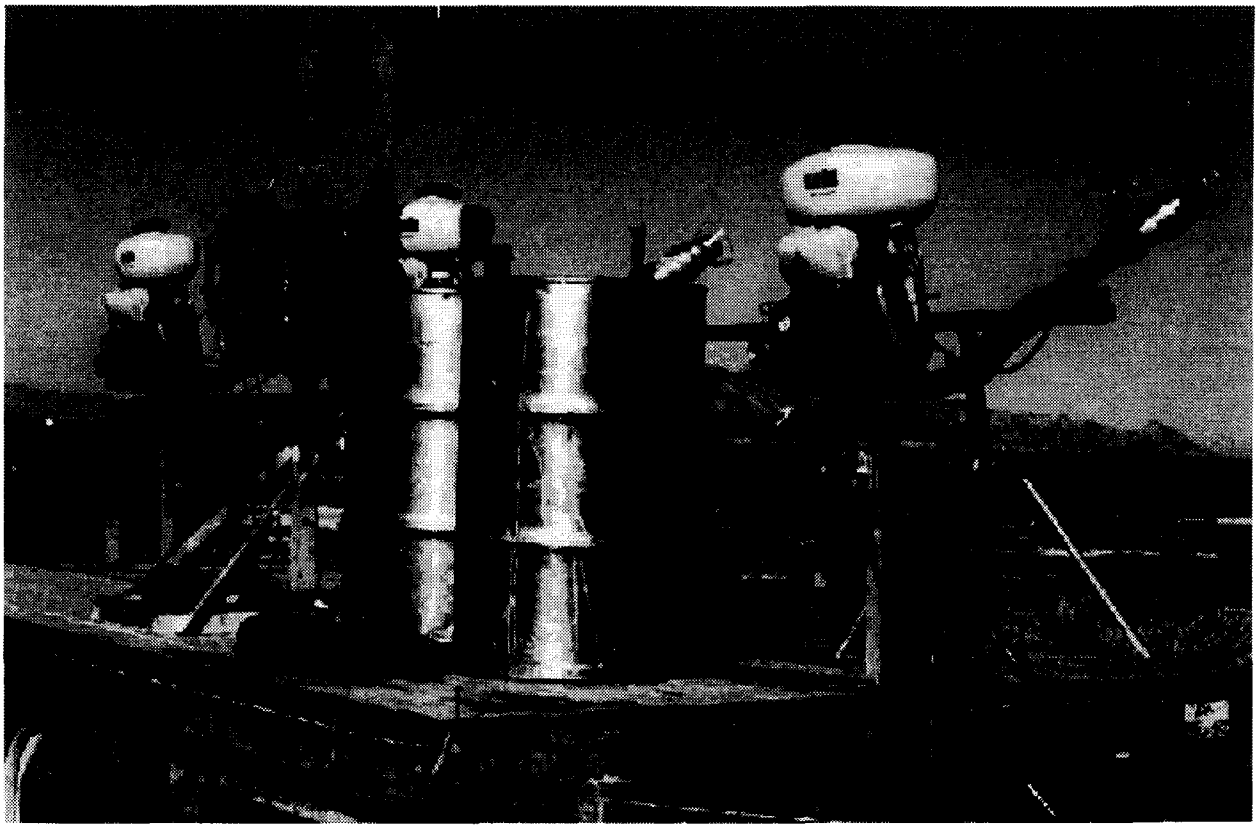


Figure A-9. Micronair Generators Mounted on a Trailer with Support Equipment Including Fifty-five Gallon Drums Containing *Bacillus subtilis* var. *niger*

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M37 Vacuum Sequencer

The M37 Wire Grid Sequential Sampler is a 110 volt, wire-controlled sampling device capable of collecting sequential samples at predetermined intervals. The control unit contains an internal vacuum distribution system with a control panel and vacuum confections. Vacuum is supplied from an external source. It can be used with AGI-30s, slit, and other samplers.

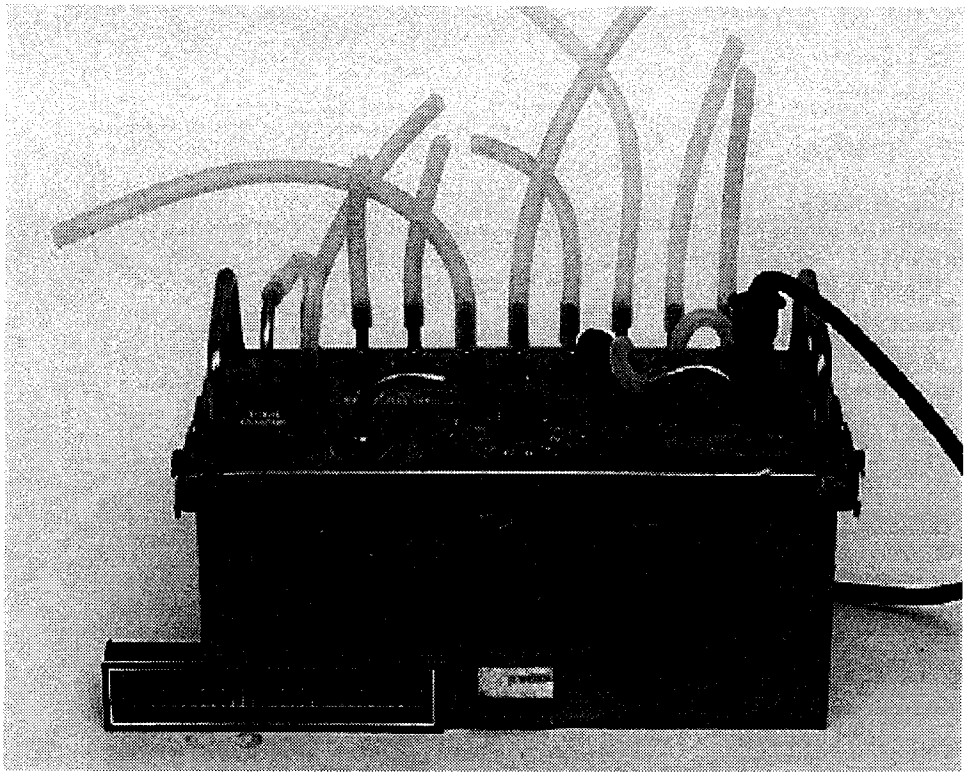


Figure A-10. M37 Vacuum Sequencer.

When assembled for operation the M37 has three main parts:

a. The control unit which contains the internal vacuum distribution system with the control panel and vacuum connections.

b. The bubbler box, stainless steel and rectangular in shape, has one row of holes into which ten AGI-30s may be placed. The tenth position is for continuous dose, and vacuum is supplied to that position starting with the first sequential sample. The bubbler box is attached to the front side of the control unit.

c. The vacuum is supplied to the M37 from an external source by means of rubber tubing connecting the M37 to either a stand alone pump (Figure A-11) or a piped vacuum source. Applying vacuum to each of the sequential AGI-30s is accomplished by energizing a separate solenoid for each individual vacuum port. The position indicator is located on the control panel and has two nonaspirating positions, No. 0 and 11. The No. 0 indicator position is the starting position. No. 10 indicates the completion of the index cycle. As the sampler is indexed to the No. 1

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position, vacuum is applied to the No. 1 bubbler and so on, sequentially. The AGI-30s are connected to the vacuum distributor outlets on the control panel by means of 3/16-in. inside diameter tubing. The AGI-30s are aspirated at 1 /pm by means of 1-/pm orifices that are installed internally in the vacuum lines.

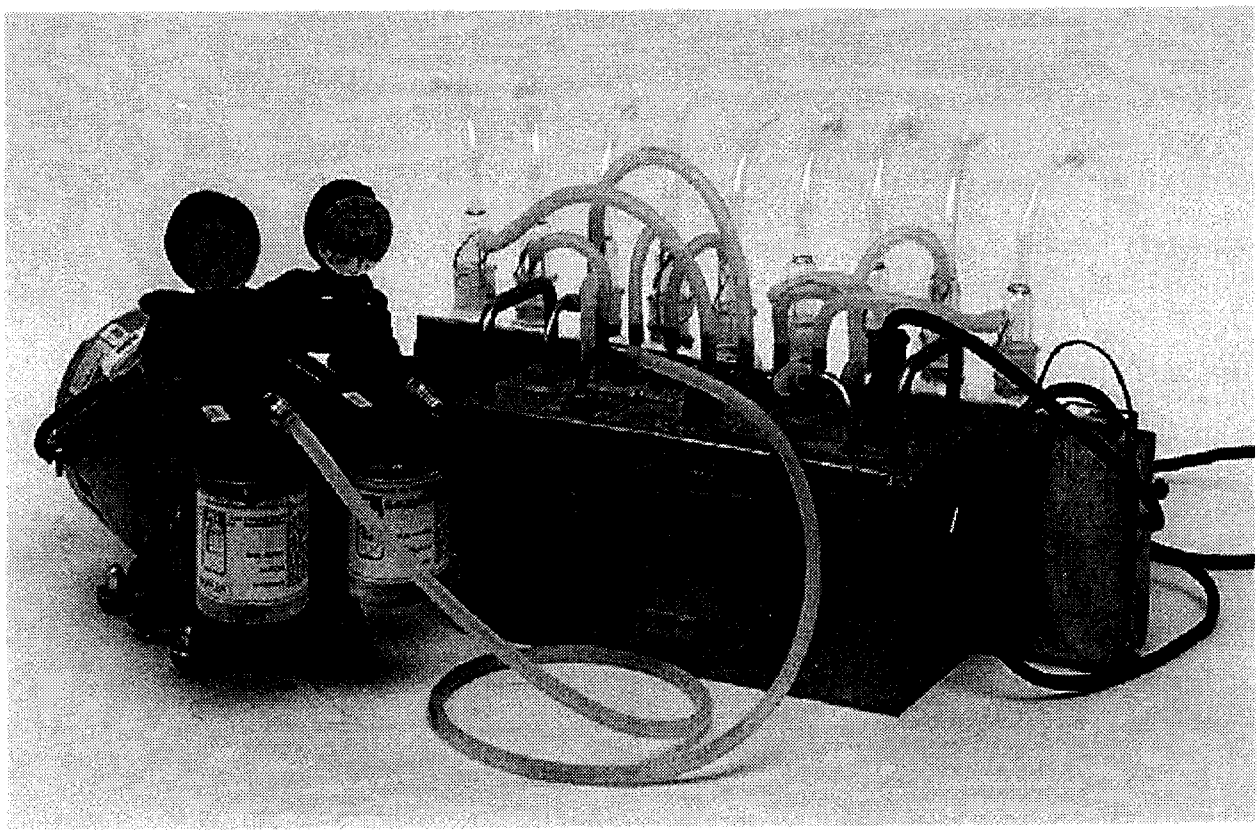


Figure A-11. M37 Vacuum Sequencer with Associated AGI-30 Sampler Rack and Electric Vacuum Pump.

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Propylene Detection System

A propylene detection system was designed and built to DPG specifications to be used as a tracer for airborne biological simulants. The system consists of eighteen Model TIP-SJ3 fast response photo ionization sensors Figure A-12), 18 radio frequency (RF) modems, a rack mounted data acquisition computer, a rack mounted calibrator, a manual propylene gas release system, mounts and sun shields for sensors and modems, and a color printer. A custom software system provides data acquisition and also controls the sensors and manages calibration. The TIP-SJ3 sensors are normally mounted at a fixed or mobile mast on tripods with RF modems. A 12 VDC Ni-Cad battery is placed near each sensor to provide power for the sensor and modem. The data acquisitions system must be within 2 Km from the farthest sensor for the radio modems to operate properly.

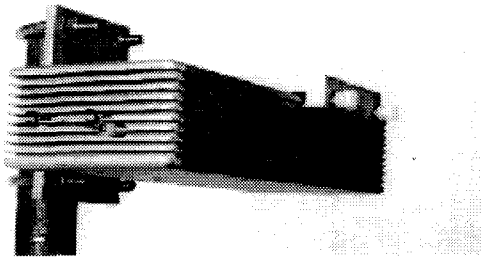


Figure A-12. Fast Response Photo Ionization Sensor.

Propylene gas is typically released at a rate of 450 to 500 *l*pm. At this concentration, propylene gas is easily detected at 800 m and can be detected up to 1200 m away. During a typical trial, propylene gas is released approximately 30 seconds before the biological simulant is aerosolized. When propylene is detected at the test area, referee sampling is initiated so that the leading edge of the simulant cloud is captured. Figure A-13 shows a number of propylene tanks on a trailer transport. The propylene tanks are placed in a small stock tank containing warm water. This facilitates gas release.

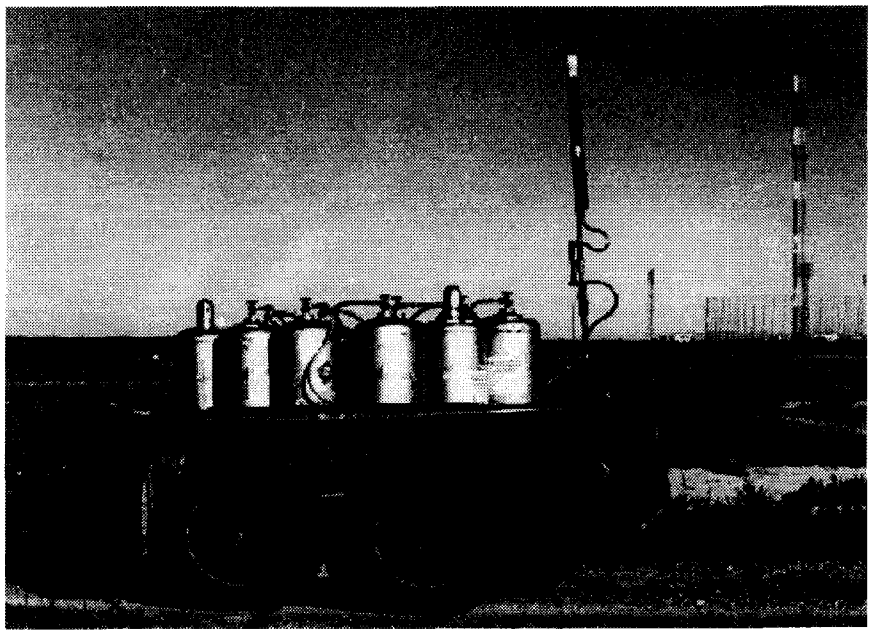


Figure A-13. Propylene Tanks in a Transport Trailer at Tower Grid.

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Chamber Aerosol Generators

1. Spinning Top Aerosol Generator (STAG)

The STAG (Figure A-14) is designed to provide a near monodispersed aerosol using an air-driven rotor. Liquid is fed through a needle, located in the center of the spinning disk, by a syringe pump. Air flow, used to rotate the top and provide suction to remove small particles, is adjusted using a separate external control box. Small satellite particles are trapped in the exhaust flow so that only the desired larger particles are expelled from the spinning surface as it rotates. Aerosol particle size is governed by; liquid feed rate of the syringe pump, height of the needle above the spinning surface, speed of rotation of the spinning disk, and air flow through the exhaust system.

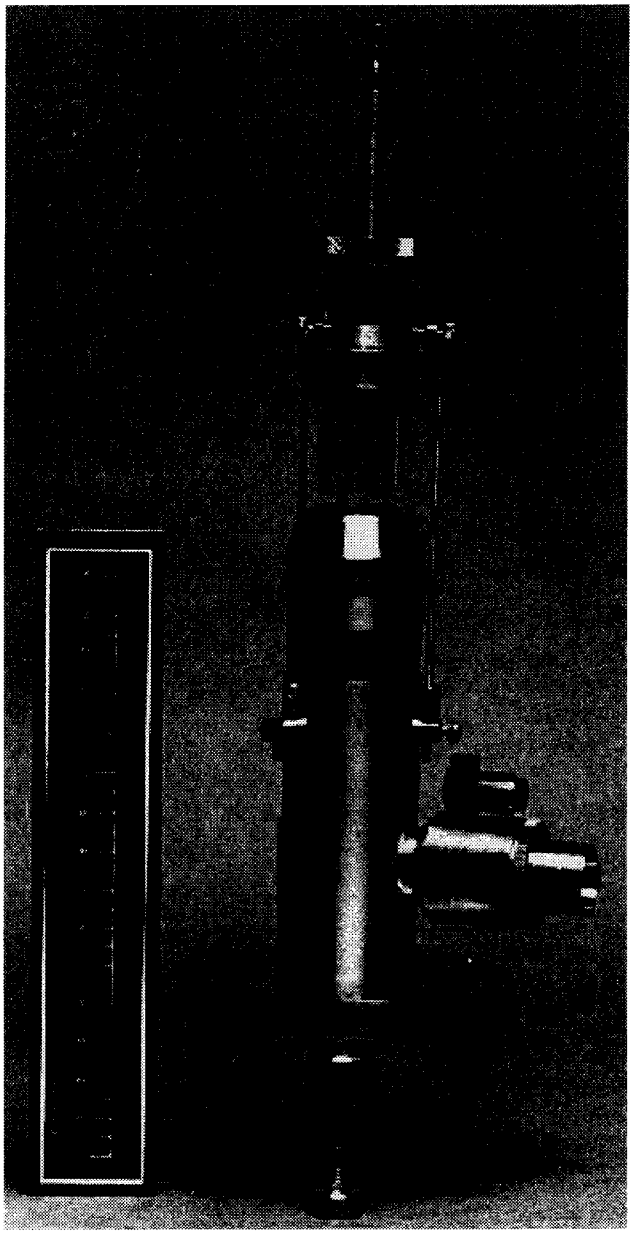


Figure A-14. Spinning Top Aerosol Generator

Successful handling of the STAG relies as much on experience as with following the manufacturers directions. The operator must be very careful to control all air pressures and flows. If back pressure is present at the exhaust manifold, satellite particles will not be drawn into the STAG and polydispersed aerosol will result. The needle height above the spinning top must be precise, or the resultant particle will be too large or will not be monodispersed. the STAG must always be used with a real time particle analyzer so that when operating parameters are altered, the resultant effect on particle size can be monitored.

2. Babington Aerosol Generator

The Babington aerosol generator (Figures A-15 and A-16) is a very simple design that has been found to produce high relative particle numbers of aerosols in the 5 to 7 μm range. The generators are manufactured by Babington engineering, McLean, Virginia, and are normally made from either glass or stainless steel. A small hollowchamber, about the size of a marble, is blown from glass and a small slit, about 1 mm in length and with varying widths, is cut into the sphere. The hollow sphere is pressurized with air (from 10 to 30 psi), and liquid is introduced at the top of the sphere with a syringe or peristaltic pump. Gravity carries the fluid over the slit as the pressurized air atomizes the liquid to produce the aerosol.

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The Babington produces polydispersed aerosols with mass median diameters ranging from submicrometer to 7 to 8 μm. The size of the resultant aerosol depends on the type of liquid used (wetting properties with glass), the size of the slit, air pressure inside the sphere, amount of liquid being dispensed, and the air baffling design of the aerosol outlet. Babington aerosol generators have proved useful when polydispersed test aerosols with high MMDs (7 μm) are required.

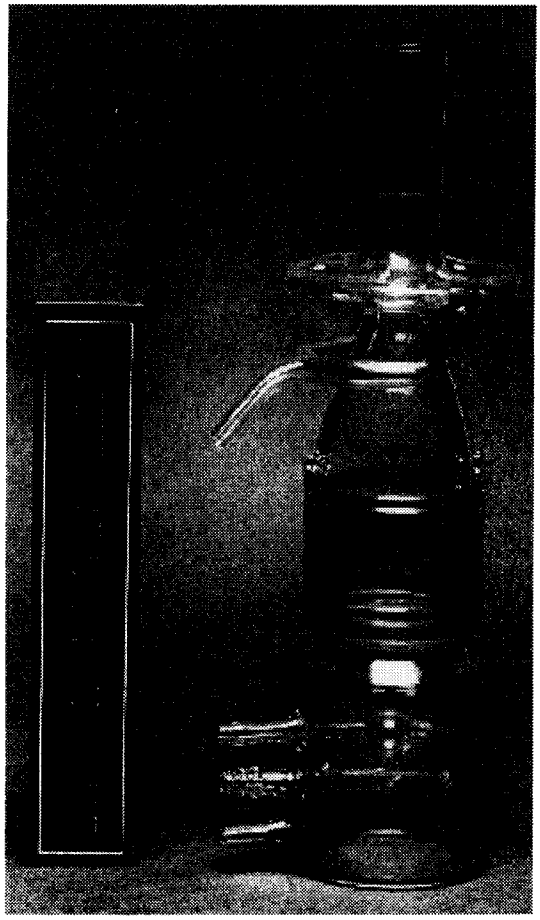


Figure A-15. Babington Generator.

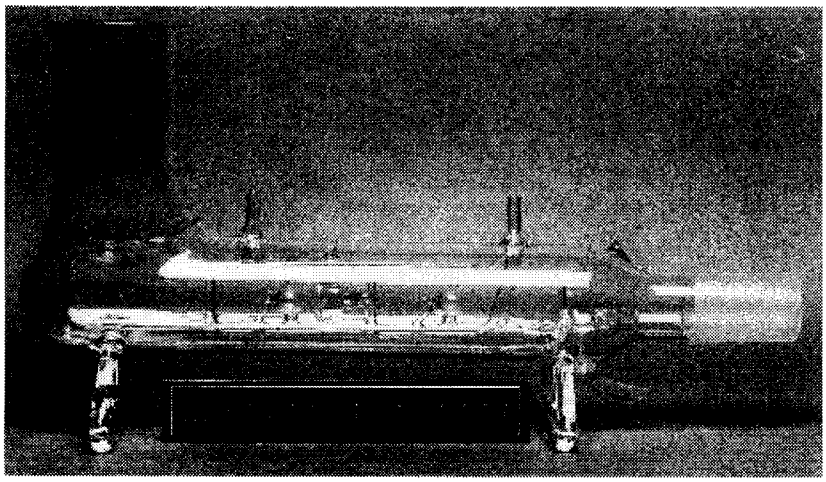


Figure A-16. Babington Generator.

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Aerosol Simulant Exposure Chamber (ASEC)

The ASEC is located in Room 28, Building 2028, Baker Laboratory, WDTC. The ASEC is an environmentally controlled chamber capable of containing simulant aerosols of BG, EH, MS2, and ovalbumin using negative pressure. The interior of the ASEC is accessible by either an airlock (4 ft.,8 in. by 4 ft., 3 in. by 9 ft.) or two panels that have two 1-in. and three 1-1/2 in. ports for passing cables and tubing through the chamber walls. Outside air is drawn into the interior of the chamber through two louvered openings equipped with HEPA filters. Exhaust air is pulled from the chamber interior through a HEPA filter before being exhausted from the building. A bank of six mixing fans, located along the south wall, ensure mixing of the simulant aerosol within the chamber.



Figure A-17. Aerosol Simulant Exposure Chamber (ASEC), Baker Laboratory, West Desert Test Center, Dugway, Utah.

The ASEC is 17 by 18 by 9 ft. The chamber walls are constructed of drywall painted with a water and UV resistant paint. Two windows, located in the north and west walls, allow users to view the interior of the chamber. The interior environment and other aspects of the ASEC are controlled using the WINMOCS® computer program. WINMOCS® is a Windows®-based program designed to control temperature, RH, negative pressure, air flow, and power inside the chamber. WINMOCS® provides two types of computer control. The first type allows the user to set parameters for negative air, inlet air flow, temperature, and RH. The computer will monitor the chamber interior environment via various probes placed throughout the interior. Using the information from the probes, the computer will turn on and off various components to maintain the parameters set by the user. The second type of computer control allows the user to have direct control of the ground fault interrupt (GFI) outlets, louvered air inlets, the bank of six mixing fans, and the valve for the inlet ports. A control window, with an on/off switch, is provided for each of

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these components in the WINMOCS® program. Specifics about the individual computer controlled components of the ASEC are as follows:

a. Temperature. Three temperature probes are located in the ceiling of the ASEC. They are located along the west and north walls and in the southeast corner of the chamber. A bank of six mixing fans is near the ceiling along the south wall. Six heating coils are located behind the mixing fans (one heating coil behind each fan). A cooling valve is located inside the mixing fan bank. The temperature range for the ASEC is 4° C to 40° C.

b. The RH. One RH sensor is located in the southeast corner of the ASEC. RH is increased using a steam release valve that is located in the bank of mixing fans. If increased RH is desired, the valve is opened and steam is released into the ASEC. The ASEC is dehumidified using a compressor that cools coils that condense moisture from the air. These coils are located inside the louvered air inlets. RH inside the ASEC is limited by the temperature inside and outside the chamber, therefore, there is no accurate range.

c. Power. Six GFI outlets are located on the interior walls throughout the chamber. These outlets are weather resistant. Power to these outlets is switched on or off by using an on/off switch in WINMOCS®.

d. Negative Pressure. Negative pressure inside the ASEC is achieved by a blower that pulls exhaust air from the chamber through a HEPA filter in the ceiling. The velocity of the air through the blower is controlled by a mass flow meter. The ASEC is capable of producing negative pressure of ½ in. of water.

e. Inlet Air. The air inlet ports are located throughout the interior of the ASEC. Inlet air is supplied to the ASEC from the building compressed air. The velocity of the inlet air is controlled by mass flow meters located in the air line.

f. Winches. There are two winches located in the ceiling of the ASEC. They are capable of lifting 100 pounds each.

The following is a description of a typical trial using the ASEC.

a. The louvers are opened and the blower is turned up to ½-in. water negative pressure. This allows a purge of the interior of the ASEC. The particle concentration inside the chamber is monitored continually using APS®. The ASEC is allowed to purge until background levels are acceptable.

b. After the interior has been purged, the louvers are closed and the blower speed is turned down. Negative pressure is always maintained during chamber use.

c. The simulant aerosol is then generated. The simulant aerosol concentration inside the ASEC is constantly monitored using the APS®.

d. When the interior aerosol concentration reaches the desired level, referee samplers are turned on using the GFI outlets.

e. After sampling is complete, the louvers are opened and the blower speed is increased to 1.2 in. of negative to purge the interior of the chamber.

f. The particle concentration in the interior of the chamber is monitored constantly. When the concentration reaches background levels, the referee samples may then be collected from the interior of the chamber. The trial is then complete.

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The ASEC was the first attempt at constructing an environmentally computer controlled chamber. The lessons learned from the building of this chamber will enhance the new chamber built in the LSTF. The computer controls for the new ASEC will not be WINMOCS® which will negate many of the limitations found with the current system.

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Biotron

The Biotron test chamber is located in Room 20, Building 2028, Baker Laboratory, WDTC. The Biotron Chamber is a BL-3 glove box that is designed for testing aerosolized BL-3 organisms and toxins. When in operation, the interior of the Biotron is accessible exclusively through an integral steam sterilizer or disinfectant dunk tank. Items being tested in the Biotron are normally housed in a Plexiglas® chamber which acts as an additional physical barrier and is connected to a mixing tube, supplied with conditioned air and an exhaust system which has two sets of redundant HEPA filters. See Table A-1 for a listing of chamber components.

The Biotron chamber is constructed of stainless steel and has eight glove ports along each side for access (Figure A-18). The chamber is 4.2 m long, 2.3 m wide, and 1.2 m high with two 71-cm by 107-cm access doors on one end (Figure A-19). The chamber is airtight and will withstand 12 cm H_2O pressure. The chamber is equipped with electrical outlets and filtered compressed breathable air. All test air is exhausted into a HEPA system which services the BL-3 suite.

The following is a description of a typical BL-3 aerosol test chamber (see Figure A-20 for location numbers).

a. Next to the metering valve (1) is a pressure sensor (2) which is connected to a photohelic vacuum gage on the master control panel. This safety feature allows monitoring of the internal air pressure with respect to the pressure outside the Plexiglas® test chamber. If the pressure inside the Plexiglas® test chamber drops below the pressure inside the Biotron chamber, compressed air to the aerosol generator will automatically be switched off.

b. A port for compressed test air (3) may be used for aerosol dilution or for purging the chamber after testing.

c. Compressed conditioned air is introduced into a 10-cm diameter stainless steel mixing tube before introduction into the Plexiglas® test chamber. Temperature and relative humidity are difficult to control inside the Biotron chamber, so most aerosol challenges are performed under ambient conditions.

d. An air-baffling arrangement (5) may be placed in the mixing tube if laminar flow is desired. The system is capable of either "pushing" compressed air or having air "pulled" through the chamber by pressure blowers at the exhaust end of the chamber.

e. Two aerosol generation ports (6) are located just downstream of the air baffling system. Their dimensions depend on the type of generator selected for the test.

f. Next is the Plexiglas® chamber which would hold the test item (8) and (9).

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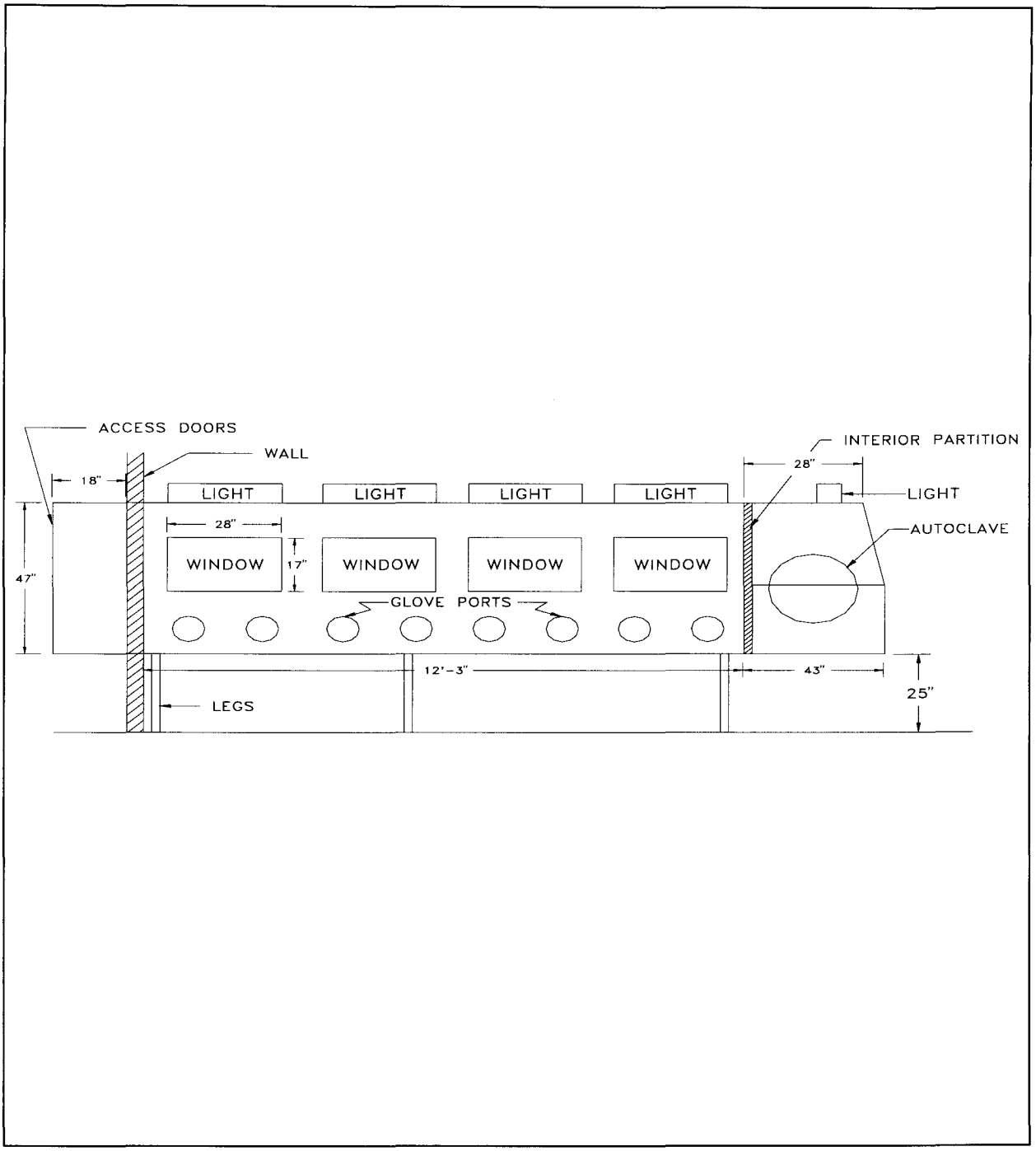


Figure A-18. Aerosol Chamber (Side View) for Biological Simulant and Agents of Biological Origin (ABO) Testing.

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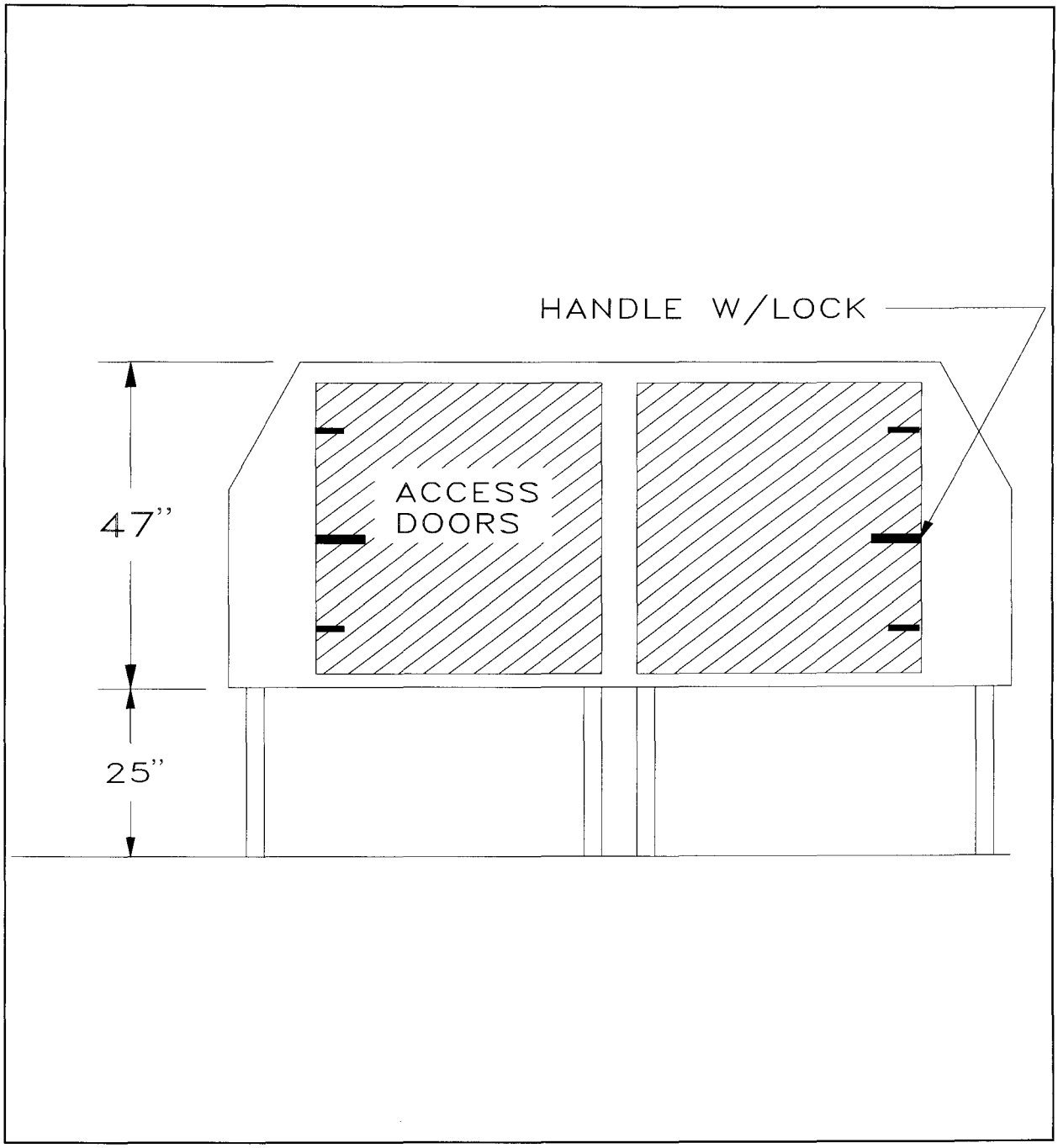


Figure A-19. Biotron Aerosol Chamber End View.

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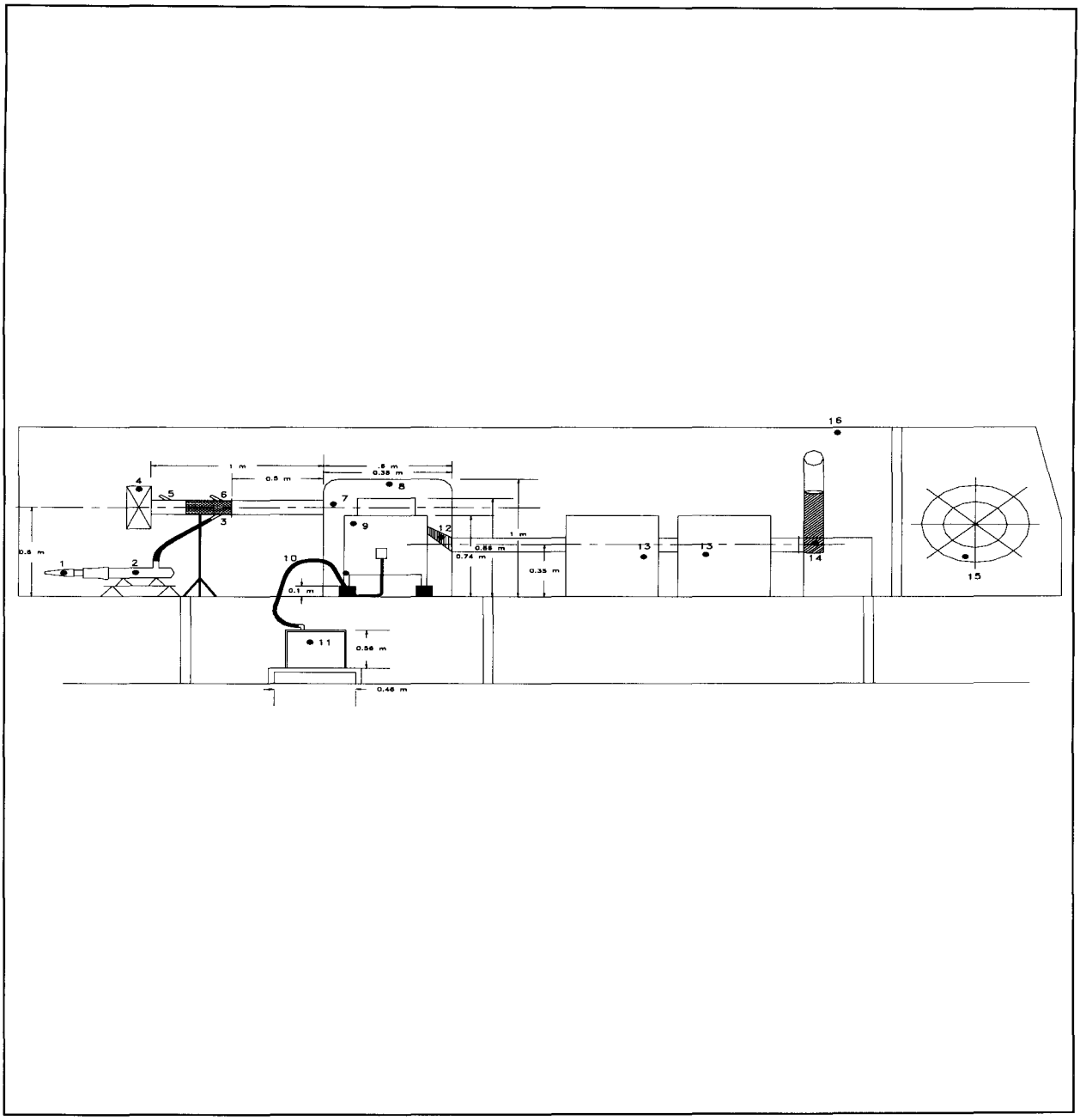


Figure A-20. Biotron Chamber with Typical Test Setup.

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Table A-1. Aerosol Chamber Components for Biological Simulant and Agents of Biological Origin (ABO) Testing.

Location[a]	Description
1	Adjustable orifice
2	Pressure gauge
3	Compressed air inlet
4	HEPA[b] filter
5	Air baffle-straightener system
6	Aerosol generation port
7	Condition sensors
8	Detector being tested
9	Detector Plexiglas® chamber
10	AGI-30[c] and aerosol characterization sample ports
11	APS®[d]
12	Aerosol return ports for the APS® and the tested detector
13	Double bag-in/bag-out HEPA filters
14	Blower fan
15	Autoclave
16	Air exhaust line with an additional two HEPA filters

[a]See Figure A-20.
[b]High-efficiency particulate air
[c]All-glass impinger, Ace Glass Company Model 30
[d]Aerodynamic Particle Sizer®

NOTE: Items not shown in Figure A-20 are master control panel with regulators, flow meters, ball valve, fan control, photo helix, differential pressure indicators, and temperature and humidity indicators and controller.

g. Several aerosol sample ports (10) are located on the Plexiglas® chamber. These sample ports are typically stainless steel tube fittings with no less than 3/8-in. inside diameter. These sample ports are used to collect a representative sample of the aerosol for characterization and viability determination. Instruments used for characterization include an APS® (a unit dedicated for Biotron use only), AGI-30 (viability determination), and Model 203 slit sampler (viability determination).

h. Contaminated challenge aerosols are exhausted through a 4-in. stainless steel tube into two high-capacity bag-in/bag-out HEPA filters. The Biotron chamber is equipped with and additional two high capacity bag-in bag-out HEPA filters through which air passes before entering the building exhaust system. If the test item is designed with a blower so that exhaust air is inherent to the test item, the exhaust air is contained and directed into the same HEPA filter system.

i. Aerosol Control Panel. The final layout of the control panel is determined by specific test objectives. A typical control panel will include the following controls. Filtered, dehumidified test air will be valved off of a main air supply line. Several compressed air lines are manifolded off the main air supply with ½ in quick disconnects. Typically, these lines are used to supply 500 *l*pm of air or less. Air flow is controlled with mass flow control devices that are scaled to the appropriate requirement. This air is then introduced into the mixing tube and acts as conditioned air or dilution air for testing. Digital controls for the mass flow meters are part of the control panel. The

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RH and temperature sensors are placed inside the Plexiglas® challenge chamber and their indicators are part of the control panel. A small, ¼-in. compressed air line is used to pressurize aerosol generation devices. A manual on/off valve and an electrically controlled valve are used to actuate the appropriate aerosol generator. Magnahelic differential pressure gauges are used to determine pressure differences between the interior of the Plexiglas® challenge chamber and the interior of the Biotron chamber. Differential pressure readings will be collected to evaluate the performance of the HEPA filters (±3 percent of reading or ±0.01 water gauge). The pressure between the interior of the Biotron chamber and the exterior of the Biotron chamber (Room 22, Baker Laboratory, WDTC) is also monitored to insure that outer areas are always positive with respect to inner areas. A calibrated hot wire anemometer is placed in the exhaust line to verify exhaust flows before introduction into the HEPA filters.

j. Test Duct Material. All materials for the test system which contact the air stream, including sample lines, are disposable or capable of being sterilized. Additionally, they are non-particle shedding, electrically conductive, and grounded.

k. Limitations connected with operating the Biotron include poor control of temperature and RH which restricts testing to ambient conditions (approximately 22°C and 35 percent RH).

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Appendix B
Facilities

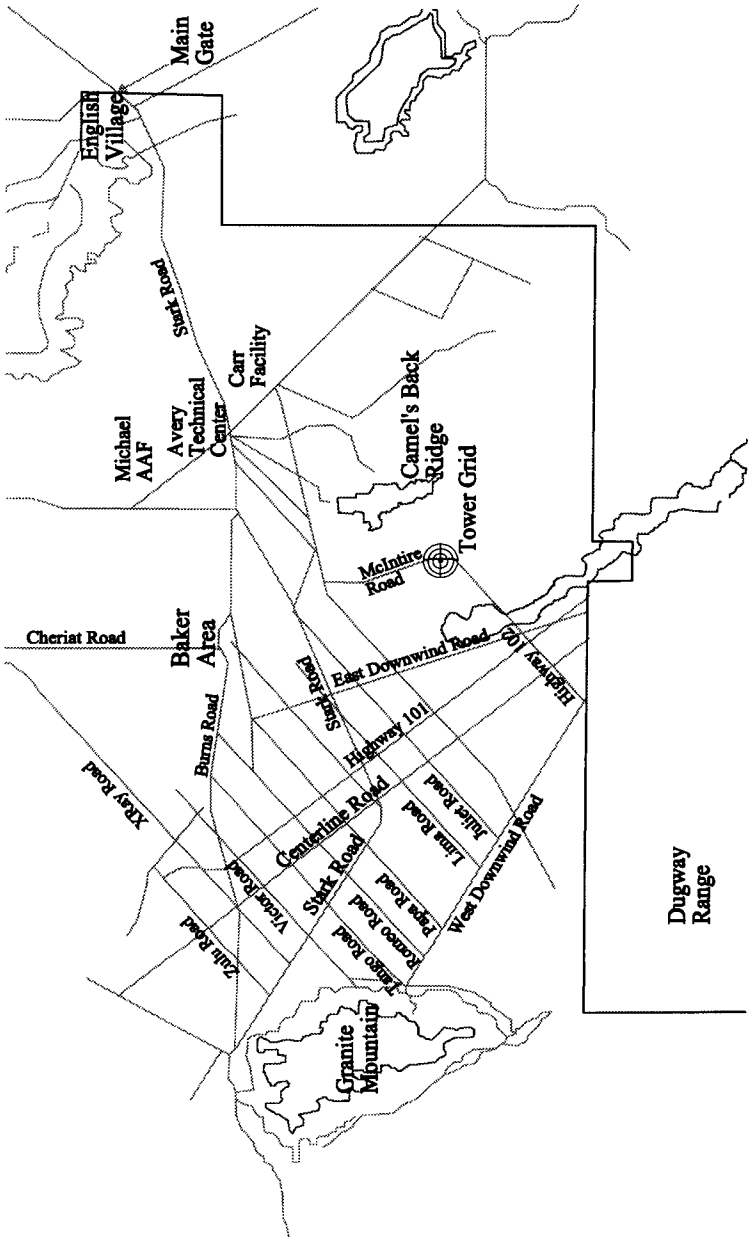


Figure B-1. Map of West Area at U.S. Army Dugway Proving Ground (DPG), Utah. Note: Tower Grid is located just west of Camel Back Ridge and Target S is located near the intersection of Hwy. 102 and Centerline Road.

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Tower Grid

Tower Grid is one of two sites that has been selected for conducting outdoor field trials at DPG. The TG is located approximately 7 miles to the southwest of the Ditto Technical Center at DPG. This site was originally selected because climatological data indicated that drainage winds are quite consistent with adequate velocities (1-8 m/sec) for aerosolizing and dispersing biological simulant aerosols. Figure B-2 is an aerial photograph of TG. The TG was originally designed to test the dispersion of chemical agents and simulants.



Figure B-2. Aerial View of the Portion of the West Area, U.S. Army Dugway Proving Ground (DPG) Covering Tower Grid (TG) and Target S. Note: Top of photograph is north.

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The inner dense array of TG is approximately 150 m in diameter (Figure B-3). Circular radials surround TG at distances of 200, 400, 800, and 1200 m, respectively, from grid center.



Figure B-3. The Central Array of Tower Grid (TG) Seen from the Northwest.

Test items are located in the inner dense array so that the sampling line is perpendicular to 160 degrees, the climatological average wind direction. Biological stimulant aerosols can be disseminated from the 200, 400, and 800 m radials but are normally generated from the 800 m arc. Because TG is round, stimulant aerosols can be generated from either the north or south ends of the grid. The limits from which test aerosols are disseminated include an arc from 130 degrees to 190 degrees on the south end, and from 310 degrees to 15 degrees on the north end. When wind directions are consistent, it is not necessary for dissemination personnel to move; the radials provide the option of moving when wind direction varies.

Anywhere from one to eight (12 m apart) referee stations can be accommodated at TG's dense array. The facilities available at TG include commercial power, fiber optics for communications and data transfer, telephone lines, a permanent command post, four large banks of lights, adequate vacuum for all referee samplers, and a gravel-covered dense array to control dust. Because of the location of Camel Back Mountain (directly east of TG), aircraft dissemination must be carefully coordinated. There are few roads located beyond the 1200 m radial; consequently, it is difficult to mark flight lines that are required for safe nighttime operations. A small asphalt-covered landing strip is located between the 800 and 1200 m radials on the north end of the grid. This landing strip is designed to support testing for unmanned airborne vehicles.

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Target S Grid

Target S is located approximately 12 miles southwest of the Ditto Technical Center. This site was originally used to study long-range dispersion of chemical agents but is now used for operational testing of biological detection systems. The large area encompassed by the grid make it an ideal location for troop movements and long-range disseminations using biological simulant aerosols. Figure B-4 is an aerial photograph of Target S. Target S is bounded on the west by West Downwind Road, on the east by East Downwind Road, and has Highway 101 and Centerline Roads running through the interior of the grid.



Figure B-4. Aerial Photograph of Target S at U.S. Army Dugway proving Ground (DPG), Utah. Note: The Dugway Range is located in the lower left corner. North is at the top of the photograph.

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On the south end of the grid, near the DPG boundary, is Baseline Road. Immediately to the northwest are a series of roads running southwest to northeast. These roads, termed the inner dense array, are spaced approximately 200 m apart. The next road to the northwest is Bravo Road and is located 800 m from the last of the inner dense array roads. Radiating outward from Bravo Road (at 800 m intervals) are Delta, Foxtrot, and Hotel Roads, respectively. Juliet is 1600 from Hotel Road, and Lima Road is 1600 m from Juliet Road.



Figure B-5. Photograph (with Telephoto Lens) of Target S Looking North from the Southern End of the Dense Array. Wildcat Mountain is in the Distance.

Sites along Delta Road have been selected when stationary sites have been desired for testing. This setup allows for biological simulant disseminations from 500 m to 5 km. Wind rose data (during July and August) indicates that the nighttime prevailing drainage winds originate from the southeast roughly 85 percent of the time. During September and October, nighttime winds will originate from either the northeast or southwest.

Advantages associated with biological detector testing at Target S include a large area for testing with few physical obstructions and the potential for testing from either the north or the south. Long range disseminations (up to 15 Km) are possible.

Disadvantages associated with testing at Target S include a lack of commercial power, no telephones or fiber optics, a poor road system (mostly dirt), and no permanent structures to support test activities. Plans are being formulated to upgrade Target S to alleviate many of the above shortcomings.

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Appendix C
Simulants and ABOs

Bacillus anthracis

1. APPLICATION.

This procedure outlines the culture and processing methods used for preparation of *B. anthracis* vegetative cells and spores in less than 10 *l* quantities for use as challenge material in testing BDA.

2. RESPONSIBILITIES.

a. The TD will:

(1) Directly manage QC aspects of *B. anthracis* spore preparation.

(2) Provide hands-on training to all technicians assigned to do *B. anthracis* spore preparation, and ensure technicians perform competently before allowing them to work independently.

b. The fermentation technician will:

(1) Follow the protocol without deviation each time the procedure is performed.

(2) Maintain comprehensive records of test data in a registered laboratory notebook.

3. SAFETY.

a. Decontamination is accomplished with 5 percent sodium hypochlorite. Sterilization is performed by steam for 1 hour at 121° C and 21 psi.

b. The human infectious dose, fiftieth percentile (ID_{50}), aerosol route is 8000 to 20,000 organisms.

c. Propagation and assay of *B. anthracis* will be performed at BL-2 in Rooms 14, 19 or 76, Building 2028, LSTF.

d. Safety Procedures:

Hazard	**Mitigation**
Incidental Aerosol	All work with viable pathogens and active toxins will be performed in either a Class II biosafety cabinet or HEPA-filtered glove box.
Access Control	All containment work will be performed in locked BL-2 or BL-3 suites.
Other	Test participants will strictly adhere to practices outlined in SOP WD-L 326, Pathogens[6] and WD-L SOP 328, Toxins[7], and DA Pamphlet 385-69, Biological Defense Safety Program[8], and Biosafety in Microbiological and Biomedical Laboratories[9].

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4. MATERIALS AND METHODS.

a. Inoculum Preparation. The media formulations used for this procedure are found in Paragraph 8. After preparation of 6 to 8 *l* of medium in a 10-*l* fermenter,

(1) Streak three to five blood agar plates (BAPs) for confluent growth from a stock slant or liquid stock and incubate overnight at 34 to 37° C. Alternatively, a liquid culture can be prepared and brought to heavy log phase growth. Between 5 and 10 percent of the fermentor expected yield should be used as an inoculum.

(2) If the BAPs contain no obvious contaminants, add the colonial material to 20 to 30 m of fermenter medium in a screw-capped tube and thoroughly suspend by vortex mixing.

(3) Examine this culture microscopically and streak a BAP for isolation to check for purity.

(4) The inoculum may be incubated prior to use at 37° C for a short time, if necessary.

b. Fermentor Setup

(1) Sterilize the desired amount of medium for 30 minutes on liquid cycle.

(2) Remove the fermenter vessel from the sterilizer and place in a laminar flow hood to cool. Use of the laminar flow hood minimizes the potential for contamination. Never cool the vessel in a cold environment such as a refrigerator or cold room because the pressure differential resulting from rapid cooling facilitates entry of contaminants.

(3) Aseptically add the salt/dextrose additive to the medium; then remove a sample for a sterility check.

(4) Set up the fermentor in a water-tight tray large enough to contain the vessel contents should the vessel or line rupture during fermentation. Use of a HEPA-filtered enclosure for the fermenter will help guard against aerosol release. Use the following checklist for setting up the fermenter:

(a) Turn on the water chiller/recirculator.

(b) Before opening the circulation valve verify correct connection of the cooling finger and condenser tower chiller lines. The chiller supply line is attached to the bottom fitting of the condensing tower.

(c) Attach the sterile-filtered air line.

(d) Check attachment and condition of the condenser tower exhaust filter(s).

(e) Insert the temperature probe.

(f) Attach the power cable to the heating element and insert it.

(g) Check the condition of the sampling port air filter and sterile sample bottle.

(h) Attach the pH probe lead (if probe is used).

(i) Attach the dissolved oxygen probe lead (if probe is used).

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(j) Connect any required sterile additives lines.

(5) Aseptically add 1 to 2 m*l* of sterile Antifoam 289 (SIGMA, St. Louis, Missouri, catalog number A-5551) to the fermenter before inoculation.

(6) Turn on the air flow to 7 to 10 *l*pm and the agitator to 500 rpm.

(7) While the vessel is under operating pressure and agitation, use 70 percent ethanol from a squirt bottle to check the headplate fittings for leaking air. If air leaks cannot be eliminated by tightening fittings, O-rings may have to be replaced. If this cannot be done aseptically, fresh medium will have to be prepared and the system resterilized and checked as described above.

(8) Turn off the air and agitator.

(9) Aseptically add the inoculum and thoroughly mix using the agitator.

(10) (Optional) Assay a sample of the fermenter culture for initial concentration and purity.

(11) Turn the air back on and turn on any monitoring or additive devices.

c. Conditions for Growth are:

(1) Maintain agitator speed at 350 to 500 rpm.

(2) The dissolved O_2 can be maintained at 50 to 60 percent with approximately 7 to 10 *l*pm air flow through the sparger. This is essential for optimal sporulation efficiency. If no oxygen probe is used, set the air flow to 7 to 10 *l*pm. The dissolved oxygen concentration will fluctuate greatly during the course of the fermentation in response to the level of biological activity.

(3) The pH should be slightly acidic but neither adjustment nor monitoring is required.

(4) The temperature should be at 34° C for a minimum of 24 hours. This temperature can be maintained for the next 24 hours, or it can be reduced to room temperature.

5. PREPARATION OF TECHNICAL OR REAGENT GRADE MATERIAL AND STORAGE.

a. Technical Grade: DPG technical grade *B. anthracis* is a wet slurry preparation prepared as follows:

(1) Harvest the fermentation product into 500 m*l*, rubber gasketed, centrifuge bottles. Centrifuge for 15 minutes at 7000 rpm at 4 to 6° C to pellet the bacteria.

(2) Working in a biosafety hood, decant the supernatant into disinfectant. Save 100 to 200 m*l* for use in pellet resuspension.

(3) Resuspend each pellet by aseptically transferring it (e.g., with a sterile spatula or pipette) to a 50 m*l* centrifuge tube. Add 15 to 25 m*l* of spent culture medium and vortex the mixture to homogeneity.

(4) Remaining vegetative cells are killed by heat treatment. The 50 m*l* centrifuge tube with the resuspended pellet is placed in a water bath at 60° C and held for 30 minutes. Treatment temperature is monitored using an identical tube containing an equal volume.

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(5) Assay the resuspended pellet. Test stock and dose concentrations are prepared in spent medium. To prevent growth of contaminants, phenol may be added to all solutions to a final concentration of 1 percent.

b. Reagent Grade. DPG reagent grade *B. anthracis* spores are prepared by washing using the following steps:

(1) After decanting the spent culture fluid following centrifugation, resuspend each pellet in sterile cold distilled water. Cold water will discourage any spores from germinating. The volume of fluid should be 50 to 100 m*l*.

(2) Centrifuge for 15 minutes at 7000 rpm at 4 to 6° C to pellet the bacteria.

(3) Decant supernatant.

(4) Repeat steps (1) to (3) at least three times, using HPLC-grade water for the final wash.

(5) Assay the resuspended pelletin HPLC-grade water. Test stock and dose concentrations are prepared in water.

c. Storage. The final pellet is resuspended, assayed, and stored frozen, or used soon. Freezing the suspension in a 50 m*l* tube inclined at about 45 to 60 degrees will allow expansion of the frozen material and reduce stress on the tube wall. Also, when shipping the material, one can determine if it was thawed and refrozen by the position of the ice in the tube. Since *B. anthracis* spores are quite hearty, it is not detrimental to immediately store the preparation at -20° C, indefinitely. If no additive is used to discourage contaminants from growing, it is advisable to store in the frozen state unless the preparation is to be used within 1 to 2 weeks. Storage at 4° C is suitable for preparations in use.

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6. QUALITY CONTROL AND QUALITY ASSURANCE

a. Identity Confirmation

(1) Colonies. On BAP, *B. anthracis* colonies are nonhemolytic, 4 to 5 mm in diameter, and off-white, with a ground-glass appearance. They are usually flat with an irregular margin and may have many comma-shaped outgrowths (Figure C-1, backlit). When the margin of the colony is pushed inward and gently lifted with an inoculating needle, the disturbed part of the colony stands up like beaten egg whites, a characteristic termed "tenacity"[c]. The vegetative cells are nonmotile (unless under starvation conditions in liquid culture medium).

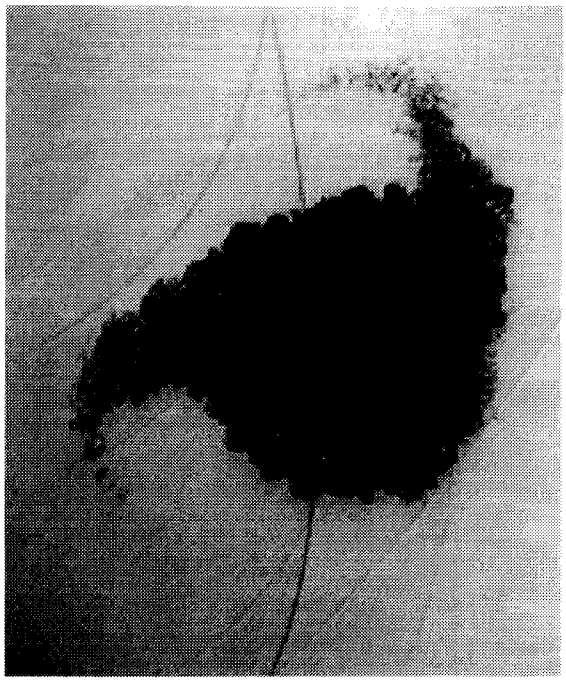


Figure C-1. *Bacillus anthracis* colony.

(2) Microscopic. The spores are centrally located and do not swell the rods. Vegetative cells are Gram's stain positive (Figure C-2). The spores are stained as follows:

(a) Prepare a heat-fixed smear.

(b) Apply a solution of 5 percent malachite green and heat to steaming for 3 to 5 minutes. Do not allow to dry. Add additional stain as required.

(c) Allow the slide to cool, and rinse off the stain with cold water until it runs clear.

(d) Counterstain with safranin dye for 20 seconds, and rinse with water until it runs clear.

(e) Free spores will appear green to greenish-blue spheres. Vegetative cells will stain pink and be rectangular or elliptical in shape (Figure C-3).

(3) Gamma phage. New strains of *B. anthracis* will be screened with gamma phage before storage and before first use in a test.

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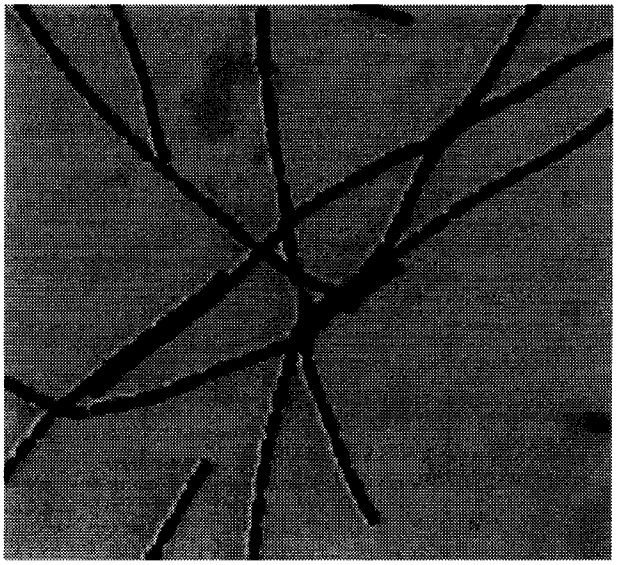


Figure C-2. *Bacillus anthracis*, Gram's stain.

(a) Prepare a lawn of the strain to be tested by inoculating 0.05 m*l* of a slightly turbid saline suspension of the organism onto a BAP and streak for confluent.

(b) Add 0.01 m*l* of the gamma phage suspension to the center of the streaked area. Let dry and invert the plate; incubate at 35° C.

(c) Repeat with known cultures of *B. anthracis* and *B. cereus* as controls.

(d) Plates should be incubated for 18 to 24 hours. A positive reaction consists of an area devoid of growth where the phage was applied, surrounded by a mat of luxuriant growth. A few resistant colonies may be seen in the phage-inoculated area[c].

b. Inactivation and Sterility Verification

(1) Inactivation. The method chosen to inactivate *B. anthracis* spores will be chosen by the customer. Despite the method chosen, the goal is to kill the spores with minimal destruction.

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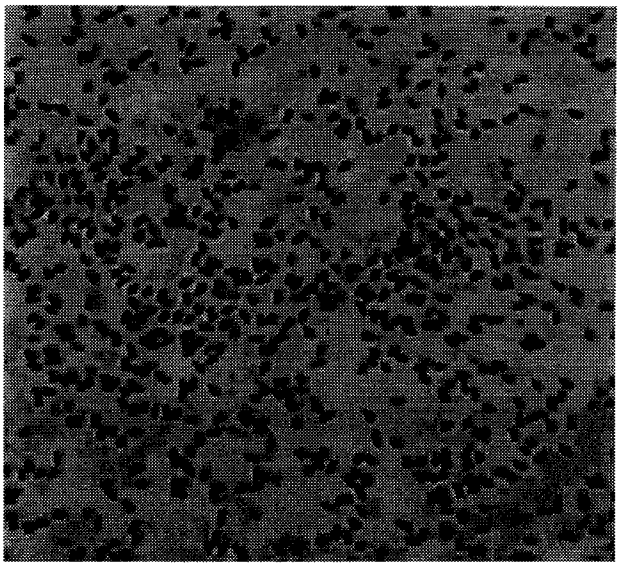


Figure C-3. *Bacillus anthracis*, spore stain

(a) Gamma irradiation. This is accomplished by exposing 30 to 40 m*l* volumes of spore slurry to 5 million rads. Figure C-4 is a scanning electron micrograph of irradiated spores.

(b) Steam. This is performed at 121° C and 21 psi in glass vessels. The length of time is dependent on volume, but 10 to 15 minutes are recommended as a starting point.

(c) Water bath. Incubate the vessel at 95° C for 30 minutes.

(2) Sterility Verification. One tenth m*l* of pelletted, inactivated spores are inoculated into 10 m*l* of brain heart infusion (BHI) medium and incubated at 37° C for 72 hours in a shaker incubator. Two hundred m*l* of this culture are inoculated onto each of three BAP and incubated for 24 hours after which they are examined for the presence of *B. anthracis* colonies.

7. DISPOSAL.

All waste culture medium and disposal materials should be steam-sterilized prior to disposal. After verification that the sterilization was successful (using *Bacillus stearothermophilus* indicators from the sterilizing run incubated at 56° C for 72 to 96 hours), the contents can be disposed of as nonhazardous refuse. Note: If using pans to contain bags, other trays, beakers, etc. while in the sterilizer, water should be placed in the pan to ensure displacement of dead, "cold" air spots and allow thorough mixing of steam throughout the load.

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8. MEDIA PREPARATIONS

a. Leighton-Doi Medium. The procedure of making this medium is referenced in two journal articles: Journal of General Microbiology, 1982, 128: 1591-1597[10] and Journal of Biological Chemistry, 1971, 246: 3189-3195[11]. This medium consists of two parts: a 2X nutrient broth base (800 m*l*) and a salt-dextrose solution (200 m*l*). Using this ratio, larger volumes of each part can be prepared.

(1) Dissolve 16 g of nutrient broth powder (Difco # 0003-17-8) in 800 m*l* of Milli Q ddH_20.

(2) Steam sterilize for 30 minutes on liquid cycle.

(3) After the medium has cooled, add 200 m of the salt-dextrose solution below for 1 *l* total volume.

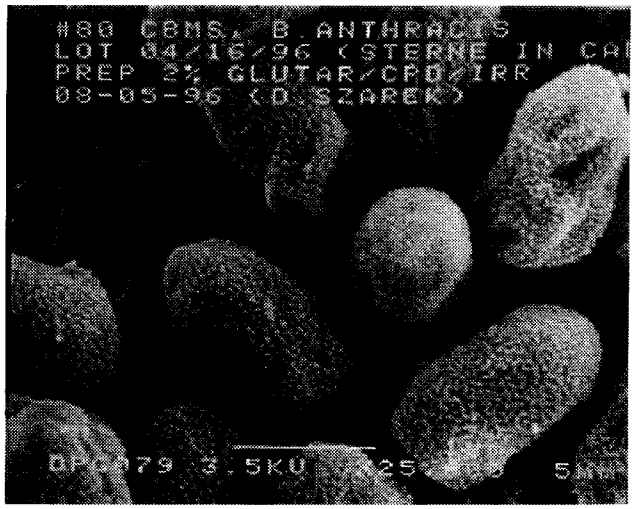


Figure C-4. *Bacillus anthracis* spores.

(4) The salt-dextrose solution consists of the following dissolved in 200 m*l* Milli Q ddH_20.

(Note: stock volumes of 1 to 2 *l* can be easily made and used as needed. This solution was filter-sterilized and aseptically added to sterile, cool nutrient broth.)

KCl	1.88 g
$CaCl_2$	0.294 g
$FeSO_4$ x $7H_2O$	0.027 g
$MnSO_4$ x H_2O	0.0169 g
$MgSO_4$ x $7H_2O$	0.0246 g
Dextrose	0.9 g

b. CAD Medium. Casein acid digest medium (CAD) is digested casein and yeast based medium. The medium consists of a casein and yeast extract base with several other ingredients added separately. The following composition makes 1 *l* of medium.

(1) Dissolve 8.5 g casein (Difco Tryptone which is a pancreatic digest of casein, No. 0123-01, or Sigma Caseine enzymatic hydrolysate, N-Z Amine A, No. C-0626) and 5 g yeast extract (Difco, 0127-01, or Sigma, No. Y-1625) in 1 *l* ddH_2O and steam sterilize for 30 minutes on liquid cycle.

(2) The remaining ingredients are made up in 5 separate stock solutions, filter-sterilized, and aseptically added to sterile, cool casein-yeast base. Stock solutions can be refrigerated; thiamine-HCl should also be stored in dark.

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Ingredient	Stock Solution	For 1 Medium
glucose (10x)	150 g/2 ddH_2O	add 10 ml
K_2HPO_4	250 g/500 m ddH_2O	add 10 ml
thiamine-HCl	1.25 g/500 m ddH_2O	add 10 ml
$CaCl_2$ (1 M)	11.1 g/100 m ddH_2O	add 0.2 ml

The following salts are combined in 100 m*l* of ddH_2O:

Ingredient	Stock Solution
$FeSO_4$ x 7 H_2O	0.1 g
$MgSO_4$ x 7 H_20	0.5 g
$MnSO_4$ x 1 H_2O	0.3 g

To each *l* of medium, add 10 m*l* of the above solution.

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9. CHALLENGE MATERIAL INFORMATION SHEET.

ORGANISM: _____

STRAIN: _____

MEDIUM: _____

DATE OF HARVEST: _____

PREPARER: _____

LAB BOOK: _____

SOP #: _____

_____ Description of how processed:_____

_____

_____ Quantity:_____

_____ Final suspension fluid:_____

_____ Assay titer:_____

_____ Phage sensitivity result:_____(+ or -)

_____ Spore stain

_____ Gram's stain

_____ Electron micrographs

_____ Gamma irradiation: date_____

_____ Sterility check: Supernatant result _____(+ or -) Slurry result_____

Project Name_____

Signature of Preparer_____

_____
QC/QA Officer

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Bacillus subtilis var. *niger*

1. BACKGROUND.

a. The *Bacillus* species spores, designated BG, and used as the primary ABO simulant for laboratory and field testing at DPG was produced during the 1960s as *Bacillus globigii* by the now-defunct Bio*ferm* Corporation of Wasco, California.

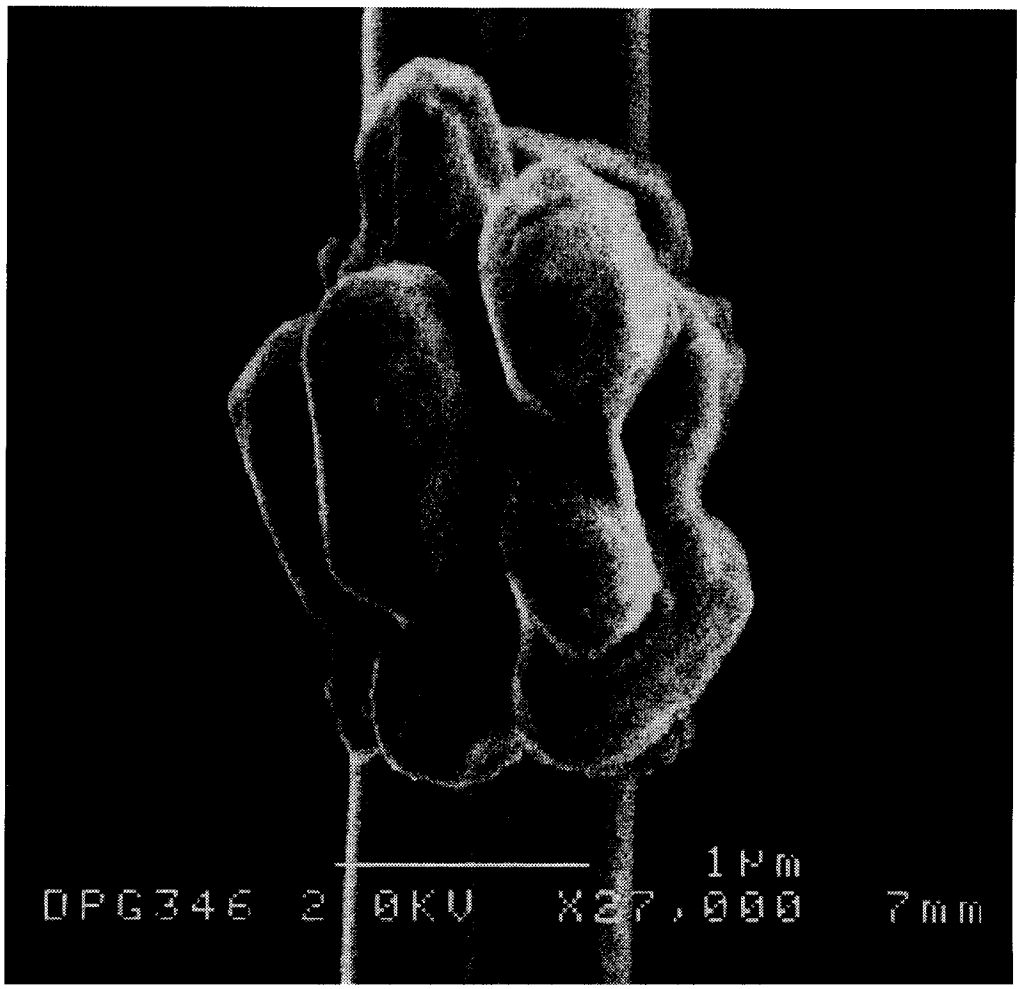


Figure C-5. *Bacillus subtilis* var. *niger* (BG) Particle Aerosolized and Captured on a Spider Web in the Aerosol Simulant Exposure Chamber.

In 1993 multiple lots of Bio*ferm* BG were sent to Pine Bluff Arsenal, Arkansas, and blended to produce 500 pounds of a single combined lot. This lot has been used since 1993 for all BDA testing requiring BG except where "freshly sporulated" BG was specified.

b. BG was produced by submerged fermentation in CAD medium or N-Z amine medium using the procedure outlined in Military Specification 50003A (CmIC), 30 June 1961[12]. The cells were collected following centrifugation into a "mud" which was twice washed with water. It was then suspended in acetone and sprayed onto an internally heated, rotating drum. This caused the acetone to be flash evaporated and left the dried BG on the

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surface of the drum. A knife edge scrapped the dried material from the rotating drum. This material was further processed in a hammer mill to achieve the desired size. Silicon dioxide was added to improve handling characteristics. Some individuals maintain that up to 20 percent may have been added. This has not been found in the literature and has not been assayed chemically.

2. CHARACTERISTICS.

BG is a Gram's stain positive rod in the vegetative state. The spores (endospores) are a nonpropagating stage highly resistant to harsh environmental conditions. Colonies are normally orange pigmented, but occasional loss of pigment is seen in cultures grown from aging slurries. BG has a unique biochemical profile as compared to "wild" *B. subtilis*. Antibodies to Bio*f*erm BG recognize surface antigens and a soluble antigen from the slurry. BG spores germinate on contact with a wide variety of liquid and semisolid media. Colonies continue to grow even at refrigerator temperatures. BG spores were selected as a simulant, in part, because of their similarity to *B. anthracis* spores, nonpathogenic nature, ubiquitous occurrence of *B. subtilis*, and unique pigmentation as compared to "wild" *B. subtilis*. Figure C-5 depicts an aerosolized BG slurry particle captured on a spider web.

3. RESPONSIBILITIES.

The assigned microbiologist will be responsible for the selection of simulant from stocks, the preparation of dilution fluids, containers, sieves and the physical assessment of the finished product. The microbiologist will prepare a simulant report stating the lot number of the stock used, the date of preparation, the viscosity, and the pH. The microbiologist will be responsible for conducting an assay of the simulant preparation to determine the active strength of the resulting preparation. The microbiologist will be responsible for overseeing the use of biological simulants in field tests to insure that material is not wasted and that unused material and empty containers are returned to the laboratory. The microbiologist in charge of simulant preparation will maintain a laboratory log book showing the following data:

a. Date prepared.

b. Lot number used.

c. Physical properties of the slurry.

d. CFU/m*l* of slurry.

e. Volume of slurry prepared.

f. Deposition of slurry.

4. SAFETY.

Proper respiratory protection, eye protection, and safety shoes will be utilized as determined by the Laboratory Safety Officer. The TD will ensure that the appropriate safety equipment is utilized by all personnel working in BG preparation IAW standard laboratory safety practices. The TD will insure that all decontamination equipment and solutions, as well as spill containment devices, are readily available prior to the start up of simulant preparation.

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5. PREPARATION.

a. Decontaminate the floor, walls, counters, and hoods in the preparation area prior to the start-up of simulant preparation using bleach solution with a minimum strength of 5.0 percent available chlorine.

b. Prepare glassware, blender flasks, and transport containers by cleaning with hot soapy water followed by a triple distilled water rinse. Steam sterilize and dry.

5.1 Dry BG.

a. Preparation. Lot strength of dry BG will be determined as follows: Prepare a sterile 99 m*l* of sterile gel phosphate with anti-foam. Weigh 1 g of dry simulant, and add this to the milk dilution bottle. Vortex mix the bottle for 5 minutes. Transfer the contents of the dilution bottle to a closed and sterile Warring® blender and blend on high speed for 30 minutes. Assessment of the resulting slurry will be done using a conventional assay with a series of 9 m*l* dilution blanks and plating each concentration onto triplicate casitone plates. The results will be the active CFU/g of dry simulant. Repeat this procedure three times for each lot number used.

b. Issue. Dry simulant is used in large quantities when supporting field tests. Issue simulant for field trials in 1-lb and 10-lb containers. Issue containers from the same lot. Record in the LSTF munitions log book the lot number and box number issued.

c. Residual material. The TD has the responsibility to ensure that unused simulant and empty simulant containers are returned to the Biotechnology Branch, Life Sciences Division, WDTC. Unused simulant will be returned to storage. Empty simulant containers will be steam-sterilized and discarded. A record of any material returned to the lab will be noted in the LSTF munitions log book.

5.2 Wet BG Slurry.

a. Preparation. Slurry will be prepared from dry simulant stock or concentrated cultures. Prepare sterile gel phosphate (with 0.10 percent anti-foam) equal to the required volume of slurry desired.

Slurries prepared from concentrated cultures require a series of dilutions derived from the assessment of the CFU/m*l* of viable cells in the culture. Enough culture stock must be prepared in advance to prepare the slurry. Slurries prepared from dry stock are simpler to prepare. Stock dry cultures are assayed to determine the CFU/g. Stock cultures usually are in the range of $1X10^{11}$ CFU/g. To prepare a slurry with $1X10^{10}$ CFU/m*l* (which is the required starting strength of slurries used in field tests), weigh 50 g of dry simulant per *l* of desired slurry. Blend with sterile gel phosphate containing anti-foam in a sterile and clean Warring® blender on high speed for 30 minutes. Filter the resulting solution through a 200 mesh sieve into a sterile and clean container.

b. Characterization. A minimum of 100 m*l* of slurry are required to perform physical property studies and viability assessment. Viability counts are reported in CFU/m*l*. Viscosity is reported in centistokes (cs).

(1) The pH. The pH of the slurry is determined using a standard pH meter. Results of this determination are recorded in the test log. Usual values are pH 6.8 to pH 7.2.

(2) Viscosity. Viscosity of the slurry is determined using a calibrated and certified viscometer. Viscosity is determined by using method American Society for Testing and Materials (ASTM) D445[13]. D445 determines the kinetic viscosity of liquids. Usual values are 1.4 to 1.9 cs. Method of time measuring for a fixed volume of liquid to flow is through the capillary of a calibrated glass viscometer under an accurately reproducible head and at a closely controlled temperature. The kinetic viscosity is calculated from the efflux time and the viscometer calibration factor.

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Water bath temperature is maintained at 15.5° C. Sample is filtered through a 200 mesh screen. Charge a clean viscometer with the sample (minimum of 60 m*l*). Condition the sample in the water bath for 60 minutes. Using a vacuum source, adjust the head level of the sample to position about 5 mm ahead of the first timing mark. Measure in seconds within 0.1 sec the time required for the meniscus to pass from the first timing mark to the second. If efflux time is less that 200 sec, select a smaller capillary viscometer and repeat the operation. Repeat this procedure 5 times. Use the average time for determining the kinetic viscosity. Record all data in the LSTF munitions log book. The formula for viscosity is: cs=Ct, where C=calibration constant for the viscometer and t=average efflux time. The equipment is required:

(a) Viscometer, capillary type, all-glass. Calibration is manufacturer certified.

(b) Thermometer, total emersion type, calibrated.

(c) The water bath is a standard water bath with mixing device for circulating water and thermostatically controlled heater.

(d) The timer is a calibrated spring-activated timing device capable of measuring to 0.20 sec or less and accuracy within 0.05 second.

(3) Specific Gravity. Specific gravity of slurry is determined by ASTM Method D422T[13].

6. RETROGRADE.

Disconnect electrical equipment from electrical power. Decontamination of equipment used in slurry preparation is accomplished by washing the equipment with copious amounts of decontamination solution with a minimum of 5 percent available chlorine. The best solution is a household bleach solution. Allow the decontamination solution to air dry. Rinse the equipment with hot water and allow to air dry for several days before reconnecting to electrical power. Glassware, disposable items, and blender containers are decontaminated by placing them in pans which contain sufficient water and soap to submerge the items. They are then steam-sterilized for 30 minutes at 21 psi. Decontaminate the slurry preparation area by washing the walls and floor with 5 percent bleach. Allow them to air dry. Rinse the walls and floor with tap water and allow them to air dry.

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APPENDIX D. ABBREVIATIONS

ABO - agents of biological origin

ACPLA - agent-containing particles per liter of air

AGI-30 - all-glass impinger

APS® - Aerodynamic Particle Sizer®

AR - U.S. Army Regulation

ASEC - aerosol simulant exposure chamber

ASTM - American Society for Testing and Materials

ATP - abbreviated test plan

BDA - biologic detector, aerosol

BG - *Bacillus subtilis* var. *niger* (*B. globigii*)

BHI - brain heart infusion

BL-3 - Biosafety Level 3

CAD - casein acid digest

CD-ROM - compact disk - read only memory

CFU - colony forming units

CP - command post

cs - centistokes

DA - Department of the Army

DPG - Dugway Proving Ground

DTP - detailed test plan

EA - environmental assessment

EH - *Erwinia herbicola*

EIS - environmental impact statement

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EMT - emergency medical technician

GFI - ground fault interrupt

HEPA - high efficiency particulate air

IAP - independent assessment plan

IAW - in accordance with

ID_{50} - infectious dose, fiftieth percentile

LOD - limit of detection

*l*PM - liters per minute

MMAD - mass median aerodynamic diameter

MNS - mission need statement

NEPA - National Environmental Policy Act

NET - new equipment training

NMAD - numerical mean aerodynamic diameter

ORD - operational requirements document

PFU - plaque-forming units

QA - quality assurance

QC - quality control

REC - record of environmental consideration

RF - radio frequency

RH - relative humidity

rpm - revolutions per minute

SAR - safety assessment report

SOP - standing operating procedure

STPM - seamless test plan manual

TD - test director

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TDS - Threat Detection Software

TECOM - U.S. Army Test and Evaluation Command

TEMP - test and evaluation master plan

TG - Tower Grid

TIC - test item chamber

TMDE - test measurement and diagnostic equipment

TOP - test operations procedure

UV - ultraviolet

WDTC - West Desert Test Center

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APPENDIX E. REFERENCES

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For Information Only

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Forward comments, recommended changes, or any pertinent data that may be of use in improving this publication to Commander, U.S. Army Test and Evaluation Command, ATTN: AMSTE-TM-T, Aberdeen Proving Ground, MD, 21005-5055. Phone: (410) 278-1486, DSN: 298-1486 and EMAIL: amstectt@apg9.apg.army.mil. Technical information may be obtained from the preparing activity: Commander, West Desert Test Center, ATTN: STEDP-WD-L, Dugway, UT, 84022-5000. Additional copies are available from the Defense Technical Information Center, 8725 John J. Kingman Road, Suite 0944, Fort Belvoir, VA 22060-6218. This document is identified by the accession number (AD. No. A328000) printed on the first page.

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